

Homological Algebra 18.715 D.G. Quillen.
Algebraic K-Theory

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(Notes by Howard Hiller)

A ring.

$K_0 A =$ Grothendieck group of fin. gen. projective A -modules.

$$K_1 A = GL_\infty(A) / [GL_\infty(A), GL_\infty(A)]$$

Look for higher K-functors $K_n A$, $n \in \mathbb{Z}$.

$K_n A = \pi_n (FA)$ $FA =$ some space associated with A .
so part of homotopy theory, so algebraic topology.

Acyclic spaces and maps:

Def: Let X be a space. X is called acyclic if $\tilde{H}_*(X) = 0$.
(integral homology). aspherical if $\pi_*(X, x_0) = 0$.

If X has homotopy type of CW complex $\Rightarrow X \sim *$ by Whitehead's Theorem.

Assumptions: All spaces have homotopy type of CW-complex, with basepoint, connected.

$$\text{Poincaré: } H_1(X) = \pi_1(X) / [\pi_1(X), \pi_1(X)].$$

Def: A group G is perfect if $G = [G, G]$ $G \rightarrow A$ (abelian) is trivial.

X acyclic $\Rightarrow \pi_1(X)$ is perfect (by Poincaré).

X acyclic and simply connected $\Rightarrow X \sim *$ (by Whitehead).

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All simple non-abelian groups are perfect; e.g. A_n $n \geq 5$.

Question: Does $\exists$ a finite acyclic polyhedron?

op & Def: Let $f: X \rightarrow Y$ be a map. TFAE:

(i) The homotopy fibre of f is acyclic.

(Replace $f: X \rightarrow Y$ by a Serre fibration, and all fibers have same homotopy type).

$$\begin{array}{ccccc} X & \xleftarrow{p} & X \times_Y Y^I & \xrightarrow{\lambda} & Y^I \\ & \searrow f & \downarrow & \searrow \lambda & \downarrow \pi \\ & & Y & \xleftarrow{\bar{\lambda}} & Y^I \end{array}$$

$Y^I = \{(x, \lambda): f(x) = \lambda(0)\}$

homotopy-fibre of $f = X \times_Y Y^I \times_{Y^I} \{y\}$

(ii) For any local coefficient system of abelian groups L on Y

$$f_*: H_g(X, f^*L) \xrightarrow{\sim} H_g(Y, L)$$

(iii) Let $\tilde{Y}$ = universal cover of Y . Then:

$$\begin{array}{ccc} X \times \tilde{Y} & \xrightarrow{p_2} & \tilde{Y} \\ \downarrow & & \downarrow \\ X & \xrightarrow{f} & Y \end{array}$$

$$(p_2)_*: H_*(X \times \tilde{Y}) \xrightarrow{\sim} H_*(\tilde{Y})$$

Proof: Assume $f: X \rightarrow Y$ is fibration and put $F = f^{-1}(\{y\})$.

(i) $\Rightarrow$ (ii) Assume F is acyclic.

Consider Serre spectral sequence:

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$$F \rightarrow X \xrightarrow{L'}$$

$$\downarrow$$

$$Y$$

$$E_{p,q}^2 = H_p(Y, H_q(F, i^* L')) \Rightarrow H_{p+q}(X, L')$$

universal coefficient theorem:

$$H_g(F, A) \cong H_g(F) \otimes A \oplus \text{Tor}_1(H_{g-1}(F), A) = 0, \quad F \text{ acyclic.}$$

Take $L' = f^* L$. Then $i^* L'$ is constant.

$$\text{Thus } H_g(F, i^* L') = \begin{cases} L & g=0 \\ 0 & g>0 \end{cases}$$

So spectral sequence degenerates, $E_{p,0}^2 \xleftarrow{\cong} H_p$, yielding (ii).

(ii) $\Rightarrow$ (iii)

$$(*) \quad \begin{array}{ccc} X \times_Y \tilde{Y} & \xrightarrow{pr_2} & \tilde{Y} \\ pr_1 \downarrow & & \downarrow p \\ X & \xrightarrow{f} & Y \end{array}$$

vertical maps
are covering spaces.

$$\begin{array}{ccc} \pi = \pi_1 Y & H_2(X \times_Y \tilde{Y}) & \xrightarrow{(p_2)_*} H_2(\tilde{Y}) \\ (p_1)_* \downarrow \cong & & \downarrow \cong \\ H_2(X, \mathbb{Z}[\pi]) & \xrightarrow{f_*} & H_2(Y, \mathbb{Z}[\pi]) \end{array}$$

canonical isomorphism
from Serre spectral sequence
for p .

(ii) $\Rightarrow f_x$ is isomorphism, so $(p_2)_*$ is.

(iii) $\Rightarrow$ (i):

$$\begin{array}{ccccc} X \times (X \times_Y \tilde{Y}) & \rightarrow & X \times_Y \tilde{Y} & \rightarrow & \tilde{Y} \\ \downarrow & & \downarrow & & \downarrow \\ X & \xrightarrow{\sim} & X' & \rightarrow & Y \end{array}$$

Look at homotopy
exact sequence.

So assume $f: X \rightarrow Y$ is fibration. But then pr_2 has same fiber as f by pull-back square $(*)$. Call it F .

Replace $f: X \rightarrow Y$ by $pr_2: X \times_Y \tilde{Y} \rightarrow \tilde{Y}$. In which case we are trying to show if $f_*: H_*(X) \xrightarrow{\cong} H_*(Y)$ and Y simply connected $\Rightarrow F$ acyclic. This is a special case of (ii) $\Rightarrow$ (i).

$$(ii) \Rightarrow (i) \quad \begin{array}{ccccc} \Omega Y & \longrightarrow & PY \times_Y X & \longrightarrow & X \\ \parallel & & \downarrow & & \downarrow f \\ \Omega Y & \longrightarrow & PY & \longrightarrow & Y \end{array}$$

Claim: $PY \times_Y X = F$, i.e. homotopy fibre of f .

Show this is acyclic; Look at spectral sequence.

$$E_{p,q}^2 = H_p(X, H_q(\Omega Y)) \Rightarrow H_{p+q}(PY \times_Y X).$$

$$E_{p,q}^2 = H_p(Y, H_q(\Omega Y)) \Rightarrow H_{p+q}(PY) \stackrel{!}{=} 0.$$

Def: F is acyclic if it satisfies conditions of Proposition.

Ex. $X \xrightarrow{\downarrow \text{trivially fibration}} *$ is acyclic $\Leftrightarrow X$ is acyclic.

(General Grothendieck method of extending properties of objects to maps)

Remarks:

(1) $X \xrightarrow{f} Y \xrightarrow{g} Z$ f, g acyclic $\Rightarrow g \circ f$ acyclic.
 $f, g \circ f$ acyclic $\Rightarrow g$ acyclic.

$$(2) \quad \begin{array}{ccc} Y' \times_Y X & \xrightarrow{g'} & X \\ \downarrow f' & & \downarrow f \\ Y & \xrightarrow{g} & Y \end{array}$$

pullback square, f, g are fibrations.
 then f acyclic $\Rightarrow f'$ acyclic.

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$$\begin{array}{ccc}
 X & \xrightarrow{f} & Y \\
 g \downarrow & & \downarrow g' \\
 Z & \xrightarrow{f'} & Z \amalg_x Y \bigg) L
 \end{array}$$

Assume f, g are cofibrations.
then f acyclic $\Rightarrow f'$ acyclic.

Pf: Assume f cofibration.

$$\begin{array}{ccc}
 \vdots & & \vdots \\
 H(X; g) & \xrightarrow{f_*} & H(Y; g') \\
 \downarrow & & \downarrow \\
 H(Z; f) & \xrightarrow{f'_*} & H(Z \amalg_x Y; L) \\
 \downarrow & & \downarrow \\
 H(Z, X; f') & \xrightarrow{f'_*} & H(Z \amalg_x Y; Y; L) \\
 \downarrow & & \downarrow \\
 \vdots & & \vdots
 \end{array}$$

f_* iso $\xRightarrow{5 \text{ Lemma}}$ f'_* is iso.

Local coefficient Whitehead Theorem:

④ $f: X \rightarrow Y$ is acyclic, $\pi_1 f: \pi_1 X \xrightarrow{\cong} \pi_1 Y$, then f is h.e.g.

Proof: Hypotheses $\Rightarrow X \times_Y \tilde{Y} =$ universal cover of $X \cong \tilde{X}$
and that $\tilde{X} \rightarrow \tilde{Y}$ induces isomorphisms of H_n .
So by simply connected Whitehead Theorem: $\tilde{X} \rightarrow \tilde{Y}$ h.e.g.

Ex: Suppose X is closed n -manifold, homology n -sphere, $n \geq 2$
 $\Rightarrow$ orientable. $H_n(X)$ has fundamental class μ



$$X \xrightarrow{f} X/X - U \cong S^n$$

$$H_n(X) \rightarrow H_n(S^n) \quad \text{maps to canonical generator}$$

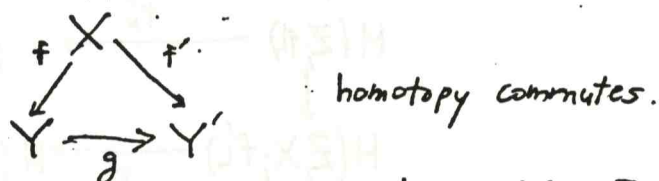
f is acyclic map, L is constant since S^n is simply-connected

Poincaré homology 3-sphere: $X = SO(3) / \text{icosahedral group} = S^3 / \text{binary icosahedral group}$
 $12 A_5$

This is a homology 3-sphere. 3-manifold with no H_1 , so no H_2 by Poincaré duality, and orientable.

Classification of acyclic with fixed source X .

Theorem: (i) If $f: X \rightarrow Y$ and $f': X \rightarrow Y'$ are acyclic and $\ker(\pi_1(f)) = \ker(\pi_1(f'))$, then $\exists$ homotopy equivalence $g: Y \rightarrow Y'$ s.t. $g \circ f \simeq f'$



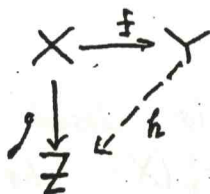
(ii) Given a perfect normal subgroup N of $\pi_1 X$, $\exists$ an acyclic map $f: X \rightarrow Y$ with $\ker(\pi_1(f)) = N$.

Remark 5: $f: X \rightarrow Y$ acyclic $\Rightarrow \pi_1 f: \pi_1 X \rightarrow \pi_1 Y$ is onto and $\ker(\pi_1(f))$ is perfect. ?

$$\pi_1 F \rightarrow \pi_1 X \xrightarrow{\pi_1 f} \pi_1 Y \rightarrow \pi_1 F$$

stronger form of (i):

Proposition: Given maps:



with f acyclic.

If $\ker(\pi_1(f)) \subseteq \ker(\pi_1(g))$. Then $\exists h: Y \rightarrow Z$ s.t. $h \circ f \simeq g$ and any two such are homotopic.

Note: Proposition $\Rightarrow$ (i) by abstract nonsense; f is initial object in maps over Z .

(7)

Proof:

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ \downarrow g & & \downarrow g' \\ Z & \xrightarrow{j'} & Z \amalg_X Y \end{array}$$

Assume f inclusion of CW complexes

Van Kampen Theorem: $\pi_1(Z \amalg_X Y) = \pi_1(Z) *_{\pi_1(X)} \pi_1(Y)$
 $= \pi_1(Z)$ by hypothesis on kernels.

f' is acyclic and $\pi_1(f')$ is isomorphism

By Remark 5, f' is homotopy equivalence.

Any h is obtained from retraction of f' , homotopy inverses are unique, so obtain uniqueness.

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classifying acyclic maps $X \rightarrow Y$, X fixed

Proof of (ii): Case $N = \pi_1 X$. Choose maps $f_i: S^1 \rightarrow X$, $i \in I$ s.t. $[f_i]$ generate $\pi_1 X$.

$$\bigvee_{i \in I} S^1 \xrightarrow{\bigvee f_i} X \rightarrow X' = \text{mapping cone of } \bigvee f_i.$$

where $X' = \bigcup_{i \in I} X \cup_{f_i} e^2 = X \cup \bigcup_{(f_i) \in I} Ve^2$

Van Kampen Theorem $\Rightarrow \pi_1 X' = 0$. For homology, have long exact mapping cone sequence.

$$0 \rightarrow H_3 X \rightarrow H_3 X' \rightarrow 0 \rightarrow H_2 X \rightarrow H_2 X' \xrightarrow{g} \bigoplus_{i \in I} \mathbb{Z} \rightarrow H_1 X.$$

$$H_1 X \cong H_1 X' \quad g \geq 3.$$

s.e.s. splits.

$$\parallel \\ 0$$

By Hurewicz theorem: $\pi_2 X' \xrightarrow{\cong} H_2 X'$, since $\pi_1 X' = 0$.

Let $g_i: S^2 \rightarrow X'$ be s.t. it maps onto i^{th} generator under

Then: $H_2 X' \cong H_2 X \oplus \bigoplus_{i \in I} \mathbb{Z}_{g_i X}(b)$ b generates H_2

$$\bigvee_{i \in I} S^2 \xrightarrow{\bigvee g_i} X' \rightarrow Y = X' \cup_{\bigvee g_i} \bigvee_I e^3 \quad \text{Then have:}$$

$$0 \rightarrow H_3 X' \xrightarrow{\cong} H_3 Y \rightarrow \bigoplus_I \mathbb{Z} \hookrightarrow H_2 X' \rightarrow H_2 X \rightarrow 0 \rightarrow \dots$$

$\begin{array}{ccc} & \uparrow & \nearrow \\ & H_2 X & \cong \end{array}$

Thus $H_2 X \cong H_2 Y \quad \forall g_i$, Y is also simply-connected by van Kampen. Finished, $X \rightarrow Y$ is acyclic with $\ker(\pi_1 f) = N$.

General case: Let X_0' = covering space of X with $\pi_1 X_0' = N$.

By first part: $X_0' \xrightarrow{f_0} Y_0$ f_0 acyclic $\pi_1 Y_0 = 0$

$$\begin{array}{ccc} X_0' & \xrightarrow{f_0} & Y_0 \\ \downarrow & & \downarrow \\ X & \xrightarrow{f} & Y \end{array} \quad \text{Let } Y \text{ be pushout.}$$

$$\text{Van Kampen} \Rightarrow \pi_1 Y = \pi_1 X \times_{\pi_1 X_0'} \pi_1 Y_0 = \pi_1 X \times_N 0 = N.$$

For fixed target Y harder problem. In particular $Y = *$.

Dror's thesis: "Acyclic spaces" M.I.T.

Remark: Any group G has a largest perfect subgroup N normal. (G is generated by all perfect subgroups)

G/N has no non-trivial perfect subgp.

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Notation: $X \rightarrow X^+$ = acyclic map killing the largest perfect subgroup of $\pi_1 X$.

(1) $X \mapsto X^+$ is a homotopy functor

(2) $(X \times Y)^+ \sim X^+ \times Y^+$

(1)

$$\begin{array}{ccc} X & \longrightarrow & Z \\ \downarrow & & \downarrow \\ X^+ & \dashrightarrow & Z^+ \end{array}$$

by universal property
on killing π_1 .

(2) Convert $X \xrightarrow{f} X^+, Y \xrightarrow{g} Y^+$ into fibrations:

$$X \times Y \xrightarrow{f \times g} X^+ \times Y^+ \text{ acyclic map.}$$

$$\pi_1 X^+ = \pi_1 X / N$$

$$\pi_1 (X^+ \times Y^+) = \pi_1 X / N \times \pi_1 Y / N'$$

$$= \pi_1 X \times \pi_1 Y / N \times N'$$

$$\pi_1 Y^+ = \pi_1 Y / N'$$

$N \times N'$ is largest perfect subgroup.

$$\text{So: } X^+ \times Y^+ = (X \times Y)^+.$$

Examples: (1) X acyclic $\Leftrightarrow X^+$ contractible (Hurewicz & Whitehead)

(2) Suppose X is a closed n -manifold with $H_*(X) = H_*(S^n)$; $n \geq 2$
(homology n -sphere). Collapsing map $X \rightarrow S^n$.

$$\text{So } X^+ \sim S^n.$$

(3) $X =$ Poincaré homology 3-sphere $= SO(3)/G = S^3/\tilde{G}$

$G =$ symmetries of the icosahedron $\approx A_5$. $|G| = 60$.

$\tilde{G} =$ binary icosahedral group; $|\tilde{G}| = 120$

$$X^+ = S^3 \text{ by (2).}$$

Take space $B\tilde{G} \simeq K(\tilde{G}, 1)$ for $\tilde{G}$ discrete.

What is $(B\tilde{G})^+$? Look at fibration.

$$\begin{array}{ccccccc} S^3 & \rightarrow & S^3/\tilde{G} & \rightarrow & B\tilde{G} & \rightarrow & BS^3 = \mathbb{CP}^\infty \\ \parallel & & \downarrow \beta & & \downarrow \alpha & & \parallel \\ S^3 & \xrightarrow[\text{degree } 120]{} & F & \xrightarrow{\gamma} & B\tilde{G}^+ & \xrightarrow{\text{induced}} & BS^{3+} \end{array}$$

$$H_*(\alpha) \text{ is } \approx \Rightarrow H_*(\beta) \text{ is iso.} \quad F = (S^3/\tilde{G})^+ = S^3.$$

Conclude: $(B\tilde{G})^+$ is a fibre space over BS^3 with fibre

Let A be an abelian group, $K(A, n)$ Eilenberg-MacLane space.

$$\pi_q(K(A, n)) = \begin{cases} A & q=n \\ 0 & q \neq n \end{cases}$$

$$\text{Hurewicz} \Rightarrow H_q(K(A, n)) = \begin{cases} 0 & q < n \\ A & q = n \\ H_{n+1} = 0 & q = n+1 \end{cases}$$

$$[X, K(A, n)] \cong H^n(X, A).$$

Lemma: Let $c: H_n(K(A, n)) \xrightarrow{\sim} A$ be canonical Hurewicz (inverse) iso. The map:

$$\begin{aligned} [X, K(A, n)] &\rightarrow \text{Hom}(H_n X, A) \\ f &\mapsto c \cdot H_n(f) \end{aligned}$$

is onto with kernel $\simeq \text{Ext}^1(H_{n-1} X, A)$.

Proof: Universal coefficient formulas

$$0 \rightarrow \text{Ext}'(H_{n-1}X, A) \rightarrow H^n(X, A) \xrightarrow{p} \text{Hom}(H_n X, A) \rightarrow 0$$

Let $X = K(A, n)$, $\exists$ unique cohomology class $u \in H^n(K(A, n), A)$ with $p(u) = c$.

Fact: $[X, K(A, n)] \xrightarrow{\sim} H^n(X, A)$
 $f \mapsto f^*(u)$

$$\begin{array}{ccc} H^n(X, A) & \xrightarrow{\quad} & \text{Hom}(H_n X, A) \\ \uparrow \scriptstyle 2 & \nearrow \scriptstyle c(H_n f) & \\ f \in [X, K(A, n)] & & \end{array}$$

Dror's construction of an acyclic space AX starting from X .

$$X_1 = X$$

$X_2 =$ the covering space of X with $\pi_1 X_2 =$ largest perfect subgroup of $\pi_1 X$.

We define inductively a tower: $\dots \rightarrow X_3 \rightarrow X_2$ etc.

$$(*) \quad \tilde{H}_g X_n = 0 \quad g < n.$$

$$(**) \quad \text{the fibre of } X_{n+1} \rightarrow X_n \text{ is } K(H_n X_n, n-1).$$

Suppose X_n is constructed: By Lemma:

$$[X_n, K(A, n)] \xrightarrow{\sim} \text{Hom}(\ast_n X_n, A).$$

Let $A = H_n(X_n)$: we get a unique map: $\chi_n: X_n \rightarrow K(H_n X_n, n)$ which induces the map:

$$\bar{c}': H_n X_n \rightarrow H_n(K(H_n X_n, n)).$$

Put $X_{n+1} = \text{fibre of } \chi_n$.

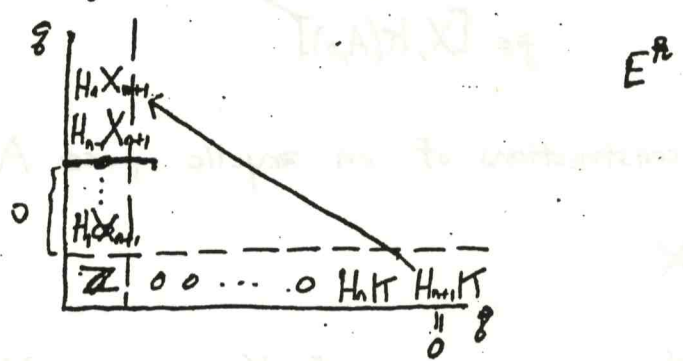
$$X_{n+1} \longrightarrow X_n$$

(**) is clear, since $\bigcap \pi(A_n) = \pi(A_{n-1})$.

$$\begin{array}{c} X_n \\ \downarrow \\ \pi(H_n X_n, n) = \pi \end{array}$$

Look at Serre spectral sequence of fibration:

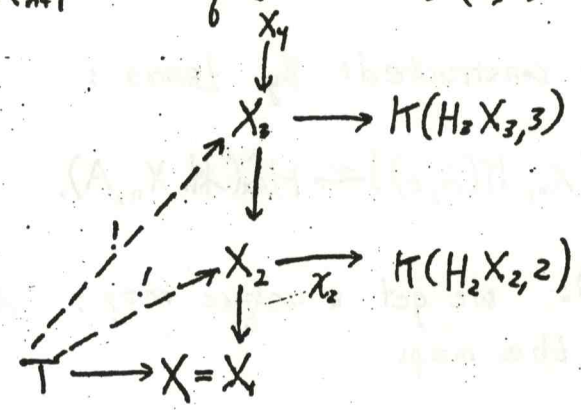
$$E_{pq}^2 = H_p(\pi, H_q X_{n+1}) \Rightarrow H_{p+q}(X_n).$$



$$\tilde{H}_q X_{n+1} = 0 \quad q < n-1$$

$0 \longrightarrow H_n X_{n+1} \longrightarrow H_n X_n \xrightarrow{\approx} H_n \pi \longrightarrow H_{n-1} X_{n+1} \longrightarrow 0$ by spectral-sequence chasing, obtain 5-term sequence.

$$\therefore \tilde{H}_q X_{n+1} = 0 \quad q < n+1 \Rightarrow (*).$$



Put $AX_\infty = \varprojlim X_n$. Then $\tilde{H}_x(AX) = 0 \therefore AX$ is acyclic

Proposition: For any acyclic space T , $[T, AX] \xrightarrow{\sim} [T, X]$.

Proof: see picture p.12. $[T, \kappa(A, n)] = H^n(T, A) = 0$.

$$\begin{array}{ccccccc} [T, \kappa(H_n X_n, n-1)] & \rightarrow & [T, X_{n+1}] & \xrightarrow{\sim} & [T, X_n] & \rightarrow & [T, \kappa(H_n X_n, n)] \\ \downarrow 0 & & & & & & \downarrow 0 \end{array}$$

Exercises: 1) Show $X^+ = \text{Cone}(AX \rightarrow X)$
2) Show $AX = \text{Fibre of } (X \rightarrow X^+)$

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Schur multipliers and central extensions:

G group:

(E_p) is a central extension of G if (by A)

$$1 \rightarrow A \rightarrow E \rightarrow G \rightarrow 1$$

and $A \subset \text{center of } E$.

(E, p) is a universal central extension of G if for any central extension (E', p') $\exists!$ homomorphism $h: E \rightarrow E'$ s.t.

$$\begin{array}{ccc} E & \xrightarrow{h} & E' \\ p \searrow & & \swarrow p' \\ & G & \end{array} \quad \text{Commutes.}$$

Proposition: (i) If (E, p) is a universal central extension then E is perfect: $E = (E, E)$.

(Hence if a universal central extension exists, G is perfect.)

(ii) Any perfect G has a universal central extension.

Let $E^{ab} = E/(E, E)$.

uniqueness
gives:

$$\begin{array}{ccc} E & \xrightarrow{(p,0)} & G \times E^{ab} \\ & \searrow (p,\pi) & \swarrow \pi \\ & G & \end{array}$$

$$\pi: E \rightarrow E^{ab} \text{ so } E^{ab} = 0 \Rightarrow E = (E, E).$$

Lemma: Let $(E, p), (E', p')$ be central extensions. If E is perfect then there exists at most one map $E \rightarrow E'$ (over G).

Proof: If $\alpha_1, \alpha_2: E \rightarrow E'$, define: $E \xrightarrow{f} \ker(p')$ by

$$f(e) = \alpha_1(e) \cdot \alpha_2(e)^{-1}. \text{ Then } f \text{ is a homomorphism.}$$

$$\alpha_1(e) = f(e) \cdot \alpha_2(e)$$

$$\alpha_1(ee') = \alpha_1(e) \cdot \alpha_1(e') = f(e) \cdot \alpha_2(e) \cdot f(e') \cdot \alpha_2(e') = f(e) \cdot f(e') \cdot \alpha_2(e) \cdot \alpha_2(e')$$

$\uparrow$
 $\ker(p')$
 $\uparrow$
 center E

Then f is a homomorphism to an abelian group $\therefore f = 0$.

$$\therefore \alpha_1 = \alpha_2.$$

Proof of (ii): Write $G = F/R$, F free. Put $E = F/(F, R)$.

Put:

$$\begin{array}{ccccccc} 1 & \rightarrow & R/(F, R) & \rightarrow & F/(F, R) & \rightarrow & F/R \rightarrow 1 \\ & & & & \parallel & & \parallel \\ & & & & E & \xrightarrow{p} & G \end{array}$$

(a) E is a central extension of G . ✓

(b) E maps to any other central extension E' of G .

$$\begin{array}{ccc} F & \xrightarrow{\alpha} & E' \\ \pi \downarrow & \nearrow & \downarrow p' \\ R/F & \xrightarrow{1} & G \end{array}$$

by freeness

$$\text{since } \alpha((F, R)) = 0.$$

(c) E is perfect.

(c) (E, E) has properties (1), (2).

Claim: (E, E) is perfect.

Take $(e_1, e_2) \in (E, E)$ $e_1 e_2 e_1^{-1} e_2^{-1}$.

Since $(E, E) \rightarrow (G, G) = G$. I know $e_i \equiv (e'_i, e''_i) \cdot a_i$ $a_i \in K$.

$(e_1, e_2) = ((e'_1, e''_1), (e'_2, e''_2))$. so: $(E, E) \subseteq ((E, E), (E, E))$

Def: If (E, p) is the universal central extension of a perfect group G , then $\ker(p)$ is called the Schur multiplier of G .

Schur: (1) Alternating group A_n , $n \geq 5$

Schur multiplier of $A_n = \mathbb{Z}/2\mathbb{Z}$.

(2) $SL_3(F_7)$ size $(2^3-1)(2^3-2)(2^3-3)$

$$7 \cdot 6 \cdot 4 = 168$$

$$PSL_2(F_7) = SL_2(F_7) / \{\pm I\} = \frac{(49-1)(49-7)}{6 \cdot 2} = 168.$$

$$SL_3(F_7) \cong PSL_2(F_7) = SL_2(F_7) / \{\pm I\}.$$

universal central extension of $PSL_2(F_7)$ is $SL_2(F_7)$.

$\therefore$ Schur multiplier is $\mathbb{Z}/2\mathbb{Z}$.

G discrete group.

BG = classifying space of topological group. (classifying principal G -bundles).

in discrete case classifies covering spaces with G acting as deck transformations.

$$G \rightarrow \tilde{B}G \xrightarrow{\text{universal covering}} \sim pt. \\ \downarrow \\ BG$$

$$BE = X_3 \xrightarrow{\partial_3} K(H_3 X_3, 3) \\ \downarrow \partial_2 \\ BN = X_2 \xrightarrow{\partial_2} K(H_2 X_2, 2)$$

Dror construction:

N = largest perfect subgroup of G .

$$BN = X_2 \xrightarrow{\partial_2} K(H_2 X_2, 2) \\ \downarrow \\ BG = X_1$$

Have fibration: $K(H_2 X_2, 1) \rightarrow X_3 \rightarrow X_2$
 $\quad \quad \quad \uparrow \quad \quad \quad \uparrow$
 $\quad \quad \quad K(N, 1) \quad \quad \quad K(N, 1)$

Put $\pi_1 X_3 = E$.

$$1 \rightarrow H_2 BN \rightarrow E \rightarrow N \rightarrow 1$$

Proposition: E is the universal central extension of N .

Proof: Recall: $F \rightarrow E \rightarrow B$ fibration of pointed spaces.
 then $\pi_1 E$ acts on $\pi_* F$ as follows.

$$\text{Loop } \alpha: [0,1] \rightarrow E \quad \text{Map } F \times [0,1] \xrightarrow{H} E \rightarrow B \\ \downarrow \quad \quad \quad \downarrow \\ [0,1] \quad \quad \quad [0,1] \\ (f, t) \mapsto p(\alpha(t))$$

$$\begin{array}{ccc} F \vee [0,1] & \xrightarrow{\quad} & E \\ \downarrow & \nearrow & \downarrow \\ F \times [0,1] & \xrightarrow{p_2} [0,1] \xrightarrow{p \cdot \alpha} & B \end{array}$$

This describes the action.

Must use homotopy extension property

$$\begin{array}{ccc}
 \pi(H_2 X_2, 1) & \xrightarrow{\cong} & \Omega \pi(H_2 X_2, 2) = \pi(H_2 X_2, 1) \\
 \downarrow & & \downarrow \\
 X_3 & \longrightarrow & * = P[\pi(H_2 X_2, 2)] \\
 \downarrow & & \downarrow \\
 X_2 & \longrightarrow & \pi(H_2 X_2, 2)
 \end{array}$$

The action of $\pi_1 X_3$ on $\pi_*(\pi(H_2 X_2, 1))$ is induced by action of $\pi_*(*) = 1$. $\therefore \pi_1 X_3$ acts trivially on $H_2 X_2 = \pi_1$ and E is a central extension of N by $H_2 X_2 \leftarrow H_2(BN)$.

Lemma: If N is perfect s.t. $H_2(BN) = 0$, then N has no non-trivial central extensions. (i.e. if $E \rightarrow N$ is a central extension, and E is perfect then $E \cong N$).

Proof: Suppose E is central extension of N .

$$1 \rightarrow A \rightarrow E \xrightarrow{p} N \rightarrow 1$$

Induces map: $BA \rightarrow BE \xrightarrow{Bp} BN$ convert Bp to fibration
 $\parallel \quad \downarrow u$ $BA = \pi(A, 1)$ is fibre.
 $K(A, 1) \rightarrow * \rightarrow K(A, 2)$ (u induces Bp).

Obstruction theory classifies fibrations with fibre $K(A, n)$ in terms of $H^{n+1}(-, A)$

Fact: Given a fibration $K(A, n) \rightarrow E \rightarrow B$ with $\pi_1(\cdot)$ acting trivially on A is induced from:

$$K(A, n) \rightarrow * \rightarrow K(A, n+1).$$

by a unique map $B \rightarrow K(A, n+1)$.

$$u \in [BN, K(A, 2)] = H^2(BN, A) = \text{Hom}(H_2 BN, A) \oplus \text{Ext}(H_1 BN, A).$$

$\downarrow$

$= 0$

so u is trivial.

This implies B_p has a section and E is trivial extension $\cong N \times A$.

$\tilde{H}_2(X_3) = 0$ $g \leq 2$ and $X_3 = BE$, we have that:

- (a) E is perfect. ($H_1(BE) = (\pi_1(BE))^{ab} = E^{ab} \neq 0$)
 (b) E has no non-trivial central extensions by Lemma.

Pf of Prop: Have shown E is a central extension of N . Show universality. Since E is perfect, it suffices to show E map to any other central extension (E', p') .

$$\begin{array}{ccccccc} 1 & \rightarrow & A' & \rightarrow & E' & \xrightarrow{p'} & N \rightarrow 1 \\ & & \parallel & & \uparrow & & \uparrow p \\ 1 & \rightarrow & A' & \rightarrow & E \times_N E' & \rightarrow & E \rightarrow 1 \end{array} \quad \text{fibre product}$$

But $E \times_N E$ is a central extension of E via pr_2 . By (b) above obtain section of pr_2 , hence a map from $E \rightarrow E$ over N .

Corollary: $H_2(BN) = \text{Schur multiplier of } N \text{ for any perfect group } N$.

$$\pi_1 X_{n+1} \rightarrow \pi_1 X_n \quad n \geq 3.$$

from fibration: $H_n(X_n, n-1) \rightarrow X_{n+1} \rightarrow X_n$.

$$\pi_1(AX) = \pi_1 X_n \quad \text{for } n \geq 3$$

$= E$ universal central extension of N

Def: (Dror): G is super-perfect if it is perfect and every central extension of G is trivial. i.e. $H_1(BG) = H_2(BG) = 0$.

Exercise: (1) Show that if X is acyclic, then $\pi_1 X$ is super-perfect.
 (2) Conversely if G is super-perfect $\exists$ acyclic space $\pi_1 X = G$.
 Let $X = A(BG)$.

Problem: Understand the functor: $X \mapsto X^+$.

e.g. Let G be finite perfect group.

Fact: $H_n(BG; \mathbb{Z})$ finite abelian groups killed by $|G|$.

Look at $(BG)^+$, a simply-connected space with

$$H_i(BG^+) = \tilde{H}_i(BG) \text{ finite, killed by } |G|$$

$$\pi_{*i}(BG^+) \text{ finite, killed by } |G|^{N_i}$$

$$\text{Homotopy theory} \Rightarrow \pi_i BG^+ = \pi_i X_p \text{ for } i \geq 1$$

can you talk about X_p in terms of groups structure of G .
 (e.g. p -subgroups of G).

9.23.74

Let A be an associative ring with 1.

Def: $K_0 A$ = Grothendieck group of f.g. projective A -modules.

$P = \text{Im}(A^n \xrightarrow{e} A^n)$ $e^2 = e$, so correspond to idempotent matrices.

Two possible defs of Grothendieck groups.

(1) Lts the ^{abelian} group having one generator $[P]$ for each fin. gen. projective P and the relations: $[P] = [Q]$ iff $P \approx Q$.
 $[P \oplus Q] = [P] + [Q]$ direct sum group

(2) Same generators $[P]$ with one relation: exact sequence Gr.

$$[P] = [P'] + [P''] \quad \text{for each s.e.s.}$$

$$0 \rightarrow P' \rightarrow P \rightarrow P'' \rightarrow 0.$$

(2) $\Rightarrow$ (1) easy. Remark: $\left(\begin{smallmatrix} \text{direct sum} \\ \text{Groth. gp} \end{smallmatrix} \right) \longrightarrow \left(\begin{smallmatrix} \text{exact sequence} \\ \text{Groth. gp} \end{smallmatrix} \right)$

and this map is iso in our case, since any s.e.s:

$$0 \rightarrow R \rightarrow E \rightarrow P \rightarrow 0 \quad \text{of } A\text{-modules}$$

with P projective splits.

Example of a category of A -modules where two Grothendieck groups differ: Let $A = \mathbb{Z}/p^2\mathbb{Z}$, take category of finitely generated A -modules. (= finite abelian groups killed by p^2). Call one M

$$M \approx (\mathbb{Z}/p\mathbb{Z})^i \times (\mathbb{Z}/p^2\mathbb{Z})^j$$

Hence $\left(\begin{smallmatrix} \text{direct sum} \\ \text{Grothendieck gp} \end{smallmatrix} \right) \approx \mathbb{Z} \oplus \mathbb{Z}$ with generators $[\mathbb{Z}/p\mathbb{Z}], [\mathbb{Z}/p^2\mathbb{Z}]$

$\left(\begin{smallmatrix} \text{exact sequence} \\ \text{Grothendieck gp} \end{smallmatrix} \right) \approx \mathbb{Z}$ with generator $[\mathbb{Z}/p\mathbb{Z}]$

use Krull-Schmidt, Jordan Hölder to give most general result.

$$0 \rightarrow \mathbb{Z}/p\mathbb{Z} \rightarrow \mathbb{Z}/p^2\mathbb{Z} \rightarrow \mathbb{Z}/p\mathbb{Z} \rightarrow 0$$

$$\text{so } [\mathbb{Z}/p^2\mathbb{Z}] = 2[\mathbb{Z}/p\mathbb{Z}]$$

Put $\mathcal{P}(A)$ = category of all fin-gen. projective A -modules.
and A -module homomorphisms

$$\text{Iso}(\mathcal{P}(A)) = \text{iso. classes of } \mathcal{P}(A)$$

This is a ...

Any abelian monoid M gives rise to an abelian $\bar{M}$ described as follows.

$$\bar{M} = M \times M / (m, s) \sim (m_1, s_1) \text{ if}$$

$$\exists t \text{ st. } m+s, +t = s+m, +t \quad (\text{equivalence relation})$$

$$+ \text{ on } \bar{M} \text{ is } (m, s) + (m_1, s_1) = (m+m_1, s+s_1).$$

$\exists$ map: $M \rightarrow \bar{M}$ given by $m \mapsto (m, 0)$ is universal

for maps of M to a group.

$$\text{Claim: } K_0 A = \overline{\text{Iso}(\mathcal{P}(A))}$$

Examples: (1) A division ring or field.

$$\text{Iso}(\mathcal{P}_A) = \mathbb{N} \Rightarrow K_0 A = \mathbb{Z}.$$

$$A = \text{PID} \quad \text{Iso}(\mathcal{P}_A) = \mathbb{N}$$

$A = \text{local ring}$ (e.g. $\mathbb{Z}/p^n \mathbb{Z}$) every fin. gen. proj is free.

$$(2) \mathcal{P}_{A \times B} = \mathcal{P}_A \times \mathcal{P}_B \text{ so } \text{Iso}(\mathcal{P}_{A \times B}) = \text{Iso}(\mathcal{P}_A) \times \text{Iso}(\mathcal{P}_B)$$

$$\text{So } K_0(A \times B) = K_0 A \times K_0 B.$$

(3) Dedekind domain A , $\text{Pic}(A) = \text{ideal class group of } A$.

$$\text{Iso}(\mathcal{P}_A) = \mathbb{N} \times \text{Pic}(A) \quad [P] \mapsto \left(\underset{f}{\text{rank}(P)}, \wedge^f P \right)$$

So $\text{Pic}(A) =$ iso classes of $P \in \mathcal{P}_A$ of rank 1.

not quite right.

$$\text{Iso}(\mathcal{P}_A) = (0,0) \cup (N - \{0\}) \times \text{Pic}(A) \subset N \times \text{Pic}(A)$$

$$[P] \mapsto (\text{rank } P, \wedge^{\text{rk}(P)} P)$$

$$\Rightarrow K_0 A = \mathbb{Z} \times \text{Pic}(A) \quad \text{gives complete description.}$$

$$(4) \quad K_0 A[T_1, \dots, T_r] = K_0 A \quad \forall \quad A \text{ regular, Noetherian} \\ (\text{e.g. } A \text{ a field}) \quad (\text{Grothendieck-Serre})$$

Not known if f.g. projective module over $F[T_1, \dots, T_r]$ is free (Serre's conjecture).

$$\text{Let } GL_n(A) = GL(n, A) = \text{group of invertible } (n \times n) \text{ matrices.} \\ = \text{Aut}(A^n)$$

$$E_n(A) = \text{subgp generated by } E_{ij}^a = I + a \begin{pmatrix} & & & j^{\text{th}} \text{ column} \\ & & 1 & \\ & & & \\ i^{\text{th}} & & & \end{pmatrix} \leftarrow i^{\text{th}} \text{ row} \\ i \neq j; \quad a \in A.$$

$$\text{Observe: (1) } e_{ij}^a e_{ij}^b = e_{ij}^{a+b}$$

$$(e_{ij}^a)^{-1} = e_{ij}^{-a}$$

$$(2) \quad (e_{ij}^a, e_{kl}^b) = e_{ij}^a e_{kl}^b e_{ij}^{-a} e_{kl}^{-b}$$

$$= 1 \quad \forall \quad \{i, j\} \cap \{k, l\} = \emptyset$$

$$\checkmark (e_{ij}^a e_{jk}^b) = e_{ik}^{a+b} \quad i, j, k \text{ distinct.}$$

$$(e_{ij}^a, e_{ki}^b) = e_{kj}^{-ba}$$

$$GL_n(A) \rightarrow GL_{n+1}(A) \\ \alpha \mapsto \begin{pmatrix} \alpha & 0 \\ 0 & 1 \end{pmatrix}$$

$\checkmark$ - gives you $E_n(A)$ perfect, for $n \geq 3$.

$$E(A) = \varinjlim E_n(A) \\ GL(A) = \varinjlim GL_n(A)$$

Whitehead Lemma: $E(A) = (E(A), E(A)) = (GL(A), GL(A)) \subseteq GL(A)$.

To show $(GL(A), GL(A)) \subseteq E(A)$.

Let $\alpha, \beta \in GL(A)$, say $\alpha, \beta \in GL_n(A)$. Work in $GL_{2n}(A)$.

$$\begin{pmatrix} \alpha\beta\alpha^{-1}\beta^{-1} & & \\ & 1 & \\ & & 1 \end{pmatrix} = \left(\begin{pmatrix} \alpha & & \\ & \alpha^{-1} & \\ & & 1 \end{pmatrix}, \begin{pmatrix} \beta & & \\ & 1 & \\ & & \beta^{-1} \end{pmatrix} \right)$$

Enough to show $\begin{pmatrix} \alpha & & \\ & \alpha^{-1} & \\ & & 1 \end{pmatrix}, \begin{pmatrix} \beta & & \\ & 1 & \\ & & \beta^{-1} \end{pmatrix} \in E_{2n}(A)$.

Proof that $\begin{pmatrix} \alpha & 0 \\ 0 & \alpha^{-1} \end{pmatrix} \in E_{2n}(A)$.

$$\textcircled{1} \begin{pmatrix} 1 & \alpha \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \alpha & 0 \\ 0 & \alpha^{-1} \end{pmatrix} = \begin{pmatrix} \alpha & 1 \\ 0 & \alpha^{-1} \end{pmatrix} \quad \textcircled{2} \begin{pmatrix} 1 & \alpha \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \alpha & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

$$\textcircled{3} \begin{pmatrix} 1 & & \\ \alpha^{-1} & 1 & \end{pmatrix} \begin{pmatrix} \alpha & 1 \\ 0 & \alpha^{-1} \end{pmatrix} = \begin{pmatrix} \alpha & 1 \\ -1 & 0 \end{pmatrix}$$

Collecting

$$\begin{pmatrix} \alpha & 0 \\ 0 & \alpha^{-1} \end{pmatrix} = \begin{pmatrix} 1 & \alpha \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ \alpha^{-1} & 1 \end{pmatrix} \begin{pmatrix} 1 & \alpha \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

$\begin{pmatrix} 1 & * \\ 0 & 1_{n-k} \end{pmatrix}$ is a product of e_{ij}^2 with $i, j \leq n-k$. etc.

Def: $K_1 A = GL(A)/E(A) = GL(A)^{ab}$.

Ex: $A = \text{skew-field } F$; Thm of Dieudonné: $GL_n(F)/E_n(F) = (F^\times)^n$.

1) $A = \text{field } F$

Thm: $(GL_n F, GL_n F) = E_n(F) = \text{matrices of determinant 1. } (SL_n(F))$
 $n \geq 1.$

(except for $n=2, F = F_2$).

$$K_1 F = F^*, \text{ since } K_1 F = GL(F)/E(F) \xrightarrow{\det} F^*$$

1') Dieudonné's theory of non-commutative determinants, for a skew field F :

$$GL_n(F)/E_n(F) \cong (F^*)^{ab} \quad \forall n \geq 1.$$

2) A Euclidean domain: $E_n A = SL_n(A) \quad \forall n \geq 2$

$$K_1 A = A^* = \text{units in } A.$$

$$K_1 \mathbb{Z} = \mathbb{Z}^* = \{\pm 1\}. \quad (\text{local rings also})$$

Def: $St(A) = \text{Steinberg group of } A$ is the group with generators x_{ij}^a , $a \in A$, $i \neq j$, $1 \leq i, j \leq n$. and the relations:

$$x_{ij}^a x_{ij}^b = x_{ij}^{a+b}$$

$$(x_{ij}^a, x_{kl}^b) = 1 \quad \text{if } i \neq l \text{ and } j \neq k.$$

$$3 \leq n \leq \infty$$

$$St_\infty(A) = St(A)$$

$$x_{il}^{ab} \quad \text{if } i, j = k, l \text{ distinct}$$

canonical homomorphism $St_n(A) \xrightarrow{i=1} E_n(A)$

$$x_{ij}^a \mapsto e_{ij}^a$$

Theorem (Milnor): $\phi: St(A) \rightarrow E(A)$ is universal central extension of $E(A)$.

Def: $K_2 A = \ker \{ \phi: St(A) \rightarrow E(A) \} = H_2(E(A))_2 = \text{Shur multiplier of } E(A)$

BG^+ ; Will show that one has following:

$$K_i A = \pi_i BGL(A)^+ \quad i=1,2. \quad (\text{not } i=0 \text{ since conned})$$

This will be our definition $i \geq 1$.

9.24.70

G group, $N =$ largest perfect subgroup.

$$X_3 = B\tilde{N} \rightarrow \pi(H_3 B\tilde{N}, 3)$$

$$X_2 = BN \rightarrow \pi(H_2 BN, 2) \quad \tilde{N} = \text{universal central extension of } N$$

$$\downarrow \alpha_1$$

$$X_1 = BG$$

tower of fibrations. $B\tilde{N} \rightarrow \pi(H_2 BN, 2)$

$$\downarrow$$

$$BG \rightarrow \pi(H_2 BN, 2)$$

$$\lim_{\leftarrow} = A(BG).$$

$$X_{n+1} \rightarrow X_n \rightarrow \pi(H_n X_n, n)$$

$$\pi_g X_{n+1} \rightarrow \pi_g X_n \rightarrow \pi_g(\pi(H_n X_n, n)) \rightarrow \pi_{g-1}(X_{n+1}).$$

$$\pi_g(B\tilde{N}) = \begin{cases} 0 & \text{ow} \\ \tilde{N} & g=1 \end{cases}$$

$$\pi_g(X_1) = \begin{cases} \tilde{N} & g=1 \\ H_3 X_3 & g=2 \\ 0 & g \geq 3 \end{cases}$$

$$\pi_g X_n = \begin{cases} \tilde{N} & g=1 \\ H_{g+1} X_{g+1} & 2 \leq g \leq n-2 \\ 0 & \text{ow} \end{cases}$$

$$\pi_g(A(BG)) = \begin{cases} \tilde{N} & g=1 \\ H_{g+1} X_{g+1} & g \geq 2. \end{cases}$$

by above result.

$$A(BG) = \text{homotopy fibre of } BG \rightarrow BG^+$$

$$ABG \rightarrow BG \rightarrow BG^+$$

$$\dots \rightarrow \pi_q A(BG) \rightarrow \begin{cases} G & q=1 \\ 0 & q \neq 1 \end{cases} \rightarrow \pi_q BG^+ \xrightarrow{\partial} \pi_{q-1}(A(BG)).$$

$$\pi_q(BG^+) = \begin{cases} G/N & q=1 \\ H_2(BN) & q=2 \\ H_q X_q & q \geq 2 \end{cases}$$

Exercise: Show the Dror tower is $AX \rightarrow \dots \rightarrow X_2 \rightarrow X_1 = X$ is the Postnikov tower for $X \rightarrow X^+$.

Take $G = GL(A)$, $E(A)$ generated by $e_{ij}^{\pm}$.

$E(A) = (E(A), E(A)) = (GL(A), GL(A))$. So $E(A)$ is largest perfect subgroup of $GL(A)$.

$St(A) =$ group with generators $x_{ij}^{\pm}$ $i \neq j$ $1 \leq i, j < \infty$. $a \in \mathbb{Z}$ and relations above.

Bass: $K_1 A = GL(A)/E(A)$

Milnor $K_2 A = \ker \{ \phi: St(A) \rightarrow E(A) \}$
 $= H_2(BE(A), \mathbb{Z}) = H_2(K(E(A), 1), \mathbb{Z}).$

$$\pi_1(BGL(A)^+) = GL(A)/E(A) \text{ by above.}$$

$$\pi_2(BGL(A)^+) = H_2(BE(A), \mathbb{Z}). \text{ So we define}$$

Def: $K_n A \simeq \pi_n(BGL(A))^+ \quad n \geq 1$

$$X_4 \downarrow$$

$$BSt(A) \longrightarrow K(H_3 BSt(A), 3)$$

Direct construction
for $BGL(A)$

$$\downarrow$$

$$BE(A) \longrightarrow K(H_2 BEA, 2)$$

$$\downarrow$$

$$BGL(A) \longrightarrow K(H, BGLA^+, 1)$$

$$K_3 A = H_3(St(A), \mathbb{Z})$$

$K_3 \mathbb{Z}$ is unknown.

($\mathbb{Z}/48$, Lie-Sapozhenko)

Universal property of $BGL(A)^+$:

$$1) \quad \begin{array}{ccc} BGL(A) & \xrightarrow{f} & BGL(A)^+ \\ & \searrow u & \nearrow \exists! h \\ & \mathbb{Z} & \end{array}$$

use obstruction theory.

If $\pi_1(u)(\pi_1 BE(A)) = 0$, then $\exists! h \Rightarrow h \cdot f \sim u$.

Recall

$$\begin{array}{ccc} X & \rightarrow & Y \\ \downarrow & & \downarrow \\ X^+ & \dashrightarrow & Y^+ \end{array}$$

$$\begin{array}{l} A \rightarrow B \text{ ring homomorphism induces} \\ GL(A) \rightarrow GL(B) \\ BGL(A)^+ \rightarrow BGL(B)^+ \\ u_* : K_n A \rightarrow K_n B. \end{array}$$

K_n is a functor from rings to abelian groups.

product of rings $A \times A'$, $GL(A \times A') = GL(A) \times GL(A')$

Fact: $B(G \times G') \sim BG \times BG'$.

$$BGL(A \times A') \sim BGL(A) \times BGL(A').$$

Fact: $(X \times Y)^+ \sim X^+ \times Y^+$

$$BGL(A \times A')^+ \sim BGL(A)^+ \times BGL(A')^+$$

Hence: $K_n(A \times A') = K_n A \times K_n A'$

Theorem: $BGL(A)^+$ is a homotopy associative and commutative H-space.

Recall $GL_n(A) \times GL_p(A) \rightarrow GL_{n+p}(A)$

$$\alpha \oplus \beta \mapsto \begin{pmatrix} \alpha & 0 \\ 0 & \beta \end{pmatrix} \quad \begin{matrix} \uparrow \\ n \\ \downarrow \end{matrix} \quad \begin{matrix} \uparrow \\ p \\ \downarrow \end{matrix} \quad \text{Whitney sum.}$$

$$\alpha \oplus (\beta \oplus \gamma) = (\alpha \oplus \beta) \oplus \gamma$$

$\alpha \oplus \beta \neq \beta \oplus \alpha$. they are conjugate in $GL_{n+p}(A)$.

Choose $N = \{1, 2, 3, \dots\}$ and choose $N \sqcup N \xrightarrow{\varepsilon} N$.

$$\varepsilon: GL(A) \times GL(A) \rightarrow GL(A)$$

$$\varepsilon_x(\alpha, \beta)_{k,l} = \text{etc.}$$

$$BGL(A)^+ \times BGL(A)^+ \rightarrow BGL(A)^+$$

Reduction of theorem to:

(29)

Lemma: Given an embedding $u: N \rightarrow M$ $N = \{1, 2, \dots\}$ then the induced map $u_*: BGL(A)^+ \rightarrow BGL(A)^+$ is homotopic to the identity.

Sublemma: u_* is a homotopy equivalence.

Lemma: Let M be the monoid of embeddings $N \hookrightarrow N$. Then any homomorphism $M \rightarrow G$ with G a group is trivial.

(2) Proof: Given any $u, v \in M$, define an embedding $v_*(u)$

$$v_*(u)(vn) = v(un)$$

$$v_*(u)(n) = n$$

$$n \notin \text{Im}(v)$$

$$\Rightarrow v_*(u) \cdot v = v \cdot u$$

Choose v so that complement of $\text{Im}(v)$ is ∞ , whence $\exists w \in M$ st. $\text{Im}(w) \cap \text{Im}(v) = \emptyset$ then $v_*(u) \cdot w = w$.

$$p(v_*(u) \cdot w) = p(w)$$

$$\parallel$$

$$p(v_*(u)) \cdot p(w) \Rightarrow p(v_*(u)) = 1.$$

but: $v_*(u) \cdot v = v \cdot u$ $p(v_*(u)) p(v) = p(v) p(u) \rightarrow p(u) = 1.$

(1) Proof: Step (1) u_* induces homology isomorphism.

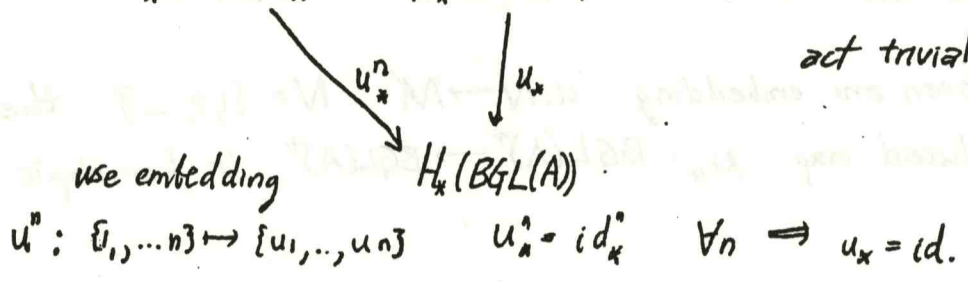
Step (2) Show $\pi_1(BGL(A)^+)$ acts trivially on $H_*(\widetilde{BGL(A)^+})$

Then can apply a suitable version of Whitehead Theorem.

$$H_*(BGL(A)^+) = H_*(BGL(A)) = \varinjlim H_*(BGL(A))$$

$$H_*(BGL_n(A)) \rightarrow H_*(BGL(A))$$

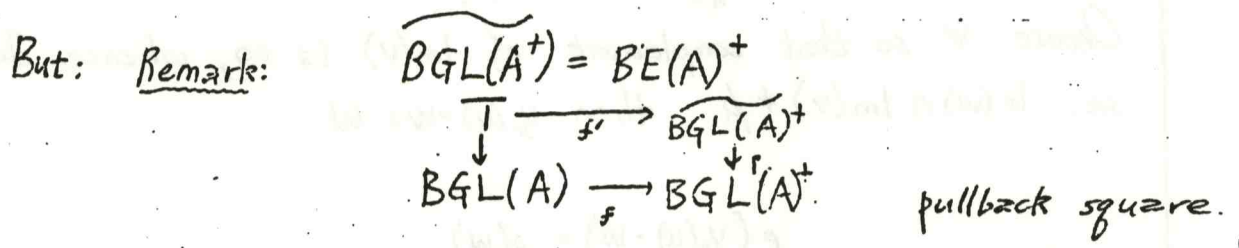
Fact: Inner automorphisms of G act trivially on $H_*(BG)$



Theorem. $BGL(A)^+$ is a homotopy commutative and associative H-sp

Key point in proof is to show $u_* \sim id$ on $BGL(A)^+$ for any embedding $u: N \hookrightarrow N$. Follows from above lemmas.

Proof of Lemma 1: By Whitehead Thm., it suffices to show u_* induces $\cong$ on $\pi_1(BGL(A)^+) = BGL(A)/E(A) = H_1(BGL(A))$ and on H_* of universal cover $\widetilde{BGL(A)^+} = BE(A)$.



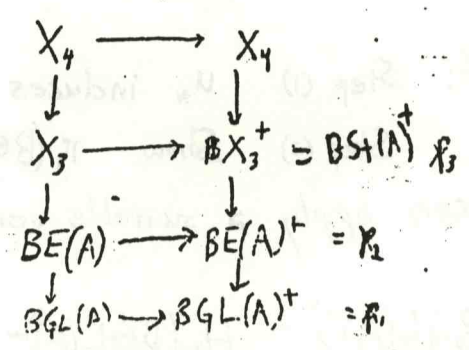
T must be the covering of $BGL(A)$ with $\pi_1(T) = E(A)$

So: $T = BE(A)$

f acyclic $\Rightarrow f'$ acyclic. so $H_*(f')$ is iso. $\Rightarrow BE(A)^T = \widetilde{BGL(A)^+}$

Exercise: Dyer tower:

Show the tower of (A) spaces is the Postnikov system of $BGL(A)^+$.



$$E(A) = \bigcup E_n A.$$

$$\Rightarrow BE(A) = \varinjlim BE_n(A) \quad \begin{array}{l} \text{converts maps to fibrations with map } \hookrightarrow \\ BE_1 \hookrightarrow BE_2 \hookrightarrow BE_3 \hookrightarrow \dots \rightarrow UB \end{array}$$

$$\Rightarrow H_*(BE(A)) = \varinjlim H_*(BE_n(A)) \quad \begin{array}{l} \text{mapping telescope.} \\ E_n = E_n(A) \end{array}$$

$$\begin{array}{ccc} E_n & \xrightarrow{i_n} & E \\ & & \downarrow u_* \\ & & E \end{array}$$

To show u_* induces identity on $H_*(BE)$ it is enough to show $u_* \circ i_n$ and i_n induce same map from $H_*(BE_n) \rightarrow H_*(BE)$.

$$\{1, \dots, n\} \mapsto \{u(1), \dots, u(n)\} \subseteq N$$

$u_* i_n$ and i_n are conjugate by an element $\sigma \in \Sigma_N \subseteq E_1 L_N$

$$\begin{pmatrix} \sigma & 0 \\ 0 & \sigma^{-1} \end{pmatrix} \in E_{2N}.$$

FACT: Conjugation acts trivially on $H_*(BG)$.

Eilenberg-MacLane group cohomology. G group, M $\mathbb{Z}[G]$ -module

$$H_*(G, M) = H_i\{P_* \otimes_{\mathbb{Z}[G]} M\} \quad \text{where}$$

$$P_* : \dots \rightarrow P_1 \rightarrow P_0 \rightarrow \mathbb{Z} \rightarrow 0$$

is a free $\mathbb{Z}[G]$ -module resolution of $\mathbb{Z}$ (trivial G -action)

$$\text{One can take } P_* = C_*(\widetilde{BG})$$

$$\begin{array}{c} \widetilde{BG} \sim 0 \\ \downarrow \\ BG \end{array}$$

$$C_i(BG, M) = C_i(\widetilde{BG} \otimes_{\mathbb{Z}[G]} M)$$

$$\text{One can show that: } C_*(BG, M) = C_*(\widetilde{BG} \otimes_{\mathbb{Z}[G]} M)$$

$$\Rightarrow H_*(BG, M) = H_*(-) = H_*(G, M)$$

Consider $(X, *) \xrightarrow{\text{evaluation at } * \in \mathbb{Z}} X^{\mathbb{Z}}$ is a fibration
 $X, \mathbb{Z}$ connected based CW

$$\begin{aligned} \pi_0[(X, *)^{(\mathbb{Z}, *)}]_{\pi_1 X} &= \pi_0(X^{\mathbb{Z}}) \\ &\parallel \\ [\mathbb{Z}, X]_{\pi_1 X} &= [Z \perp_{pt}, X] \text{ (gives homotopy class)} \end{aligned}$$

$\pi_1 X$ acts on $[\mathbb{Z}, X]$

Suppose $X = BG$; $[\mathbb{Z}, BG] \rightarrow \text{Hom}(\pi_1 \mathbb{Z}, G)$ obstruction theory

$$[Z \perp_{pt}, BG] \xrightarrow{\sim} \text{Hom}(\pi_1 \mathbb{Z}, G) / \text{conjugation by elts of } G$$

by above remarks.

An inner automorphism of G induces a map on BG which is homotopic to the identity, (not base pt preserving)
 σ induces id on $H_*(BG)$ $\varphi = D$

Theorem (Milnor & Moore): Suppose M is a connected H -space.

$$Q \otimes_{\mathbb{Z}} \pi_* M \xrightarrow{\sim} \text{Prim}\{H_*(M, Q)\}$$

$$\{z \in H_*(M, Q) : H_*(M, Q) \xrightarrow{z} H_*(M \wedge M, Q)\}$$

$$\Delta z = z \otimes 1 + 1 \otimes z$$

$$H_*(M, Q) \cong [\text{Syn alg on } \pi_{ev}(M) \otimes Q] \otimes [\text{Ext alg. on } \pi_{odd}(M) \otimes Q]$$

Canonical
algebra iso

if M is homotopy commutative and associative

See appendix to Milnor-Moore paper on "Hopf Algebras."

Take $M = BGL(A)^+$

Corollary: $H_i A \otimes \mathbb{Q} \cong \text{Prim } H_i(BGL(A), \mathbb{Q})$.

$\downarrow^2$

Pf: $\pi_i (BGL(A)^+) \otimes \mathbb{Q} = \text{Prim } (H_i(BGL(A)^+, \mathbb{Q}))$.

$H_*(BGL(\mathbb{Z}), \mathbb{Q})$ unknown. (?)

Theorem of Borel: F number field with r_1 real, r_2 complex absolute values.

$$[F:\mathbb{Q}] = r_1 + 2r_2$$

$$F \otimes_{\mathbb{Q}} \mathbb{R} \cong \mathbb{R}^{r_1} \times \mathbb{C}^{r_2}$$

Let $A =$ ring of integers in $F = \text{int closure}_F(\mathbb{Z})$.

$$\dim(H_i A \otimes \mathbb{Q}) = \begin{cases} 1 & i=0 \\ r_1 + r_2 - 1 & i=1 \\ 0 & i \equiv 2 \pmod{4} \\ r_2 & i \equiv 3 \pmod{4} \\ 0 & i \equiv 0 \pmod{4} \\ r_1 + r_2 & i \equiv 1 \pmod{4} \end{cases}$$

$$K_0 A = \mathbb{Z} \oplus P_{ic}(A)$$

$\downarrow$
finite ideal
class group

Ex 1: $A = \mathbb{Z}$

$$K_0 \mathbb{Z} = \mathbb{Z}$$

$$K_1 \mathbb{Z} = \mathbb{Z}/2\mathbb{Z} \quad (\text{units})$$

$$K_2 \mathbb{Z} = \mathbb{Z}/2\mathbb{Z} \quad (\text{Milnor})$$

$$\dim(K_3 \mathbb{Z} \otimes \mathbb{Q}) = 0 = r_2$$

$$\dim(K_4 \mathbb{Z} \otimes \mathbb{Q}) = 0$$

$$\dim(K_5 \mathbb{Z} \otimes \mathbb{Q}) = 1$$

Ex 2: $F = \mathbb{Q}[\sqrt{d}]$

$$1, 1, 0002, 0002, \dots$$

$$r_1 = 2$$

$$r_2 = 0$$

$$F = \mathbb{Q}[\sqrt{-d}]$$

$$1, 0, 0101, 0101, \dots$$

$$r_1 = 0$$

$$r_2 = 1$$

Let A be a fixed ring, $GL_n = GL_n(A)$, n fixed.

$$G_n = \left\{ \begin{pmatrix} 1_n & * \\ 0 & GL_n \end{pmatrix} \right\} \subseteq GL_{n+n}(A)$$

= group of automorphisms of exact sequences:

$$0 \rightarrow A^r \rightarrow A^{r+n} \rightarrow A^n \rightarrow 0$$

which identity on A^r
 subspace quotient A^r .

$$G_n = GL_n(A) \ltimes \text{Hom}(A^n, A^r) \longleftrightarrow GL_n(A)$$

$$G_n \xrightarrow{p} GL_n(A) \quad s(\alpha) = \begin{pmatrix} 1 & 0 \\ 0 & \alpha \end{pmatrix}$$

$$p \begin{pmatrix} 1 & * \\ 0 & \alpha \end{pmatrix} = \alpha.$$

$$H_*(G_n) = H_*(GL_n) \oplus [\ker(p_*)]$$

Theorem: $\lim_{n \rightarrow \infty} H_*(G_n) \cong \varinjlim_n H_*(GL_n)$.

Topological analogue: take $GL_n(\mathbb{C})$ with its topology.

$$B(G_n, \mathbb{C}) \xleftarrow{Bs} BGL_n(\mathbb{C})$$

$\rightarrow$ infinite dimensional

$$H \subseteq G$$

fibration:

$$G/H \rightarrow BH \rightarrow BG$$

$$\begin{pmatrix} 1 & M_{rn}(\mathbb{C}) \\ & GL_n(\mathbb{C}) \end{pmatrix} / GL_n(\mathbb{C}) \cong M_{rn}(\mathbb{C}) \quad \text{contractible.}$$

Corresponds to statement in vector bundle theory that:

short exact sequences of vector bundles split, use Riemannian metric in standard way.

Example: $A = F_p$. $GL_1 = F_p^* \subseteq \begin{pmatrix} 1 & F_p \\ 0 & F_p^* \end{pmatrix} = G_1$

$H_*(F_p^*)$ has no p -torsion

$H_0(G_1)$ has p -torsion for ∞ many n .

Show theorem needs $\lim$ above.

Remarks:

$$GL_p \times GL_q \rightarrow GL_{p+q}$$

$$(\alpha, \beta) \mapsto \begin{pmatrix} \alpha & 0 \\ 0 & \beta \end{pmatrix}$$

Whitney sum

associative, commutative up to conjugacy.

$$\begin{pmatrix} p & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \beta & 0 \\ 0 & \alpha \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$\oplus$ induces products: $H_*(GL_p) \otimes H_*(GL_q) \rightarrow H_*(GL_{p+q})$

canonical.

$M_{p,q}$

$\oplus$

$$H_*(GL_p) \otimes H_*(GL_q) \xrightarrow{M_{p,q}} H_*(GL_{p+q})$$

$$H_*(GL_{p+q}) \xrightarrow{M_{p,q}} H_*(GL_{p+q})$$

one direction

other

$$\begin{pmatrix} \alpha & \beta \\ 0 & \gamma \end{pmatrix}$$

conjugate.

aligning.

One gets a product in

$H_*(GL) = \varinjlim H_*(G_i)$, get ring structure on Λ -algebra, as above

bc is commutative
 $H_*(X) = H_*(X, \Lambda)$

$\approx H_*(BGL^+)$
 $\sim$ Postnikov
provides
for the
 H -space.

Return to Theorem: (homology with coefficients in Λ .)

Proof: Reduce to case where $\Lambda = \mathbb{F}_p, \mathbb{Q}$.

(Consider the category $\mathcal{C}$ of abelian groups for which this holds.)

$$0 \rightarrow \Lambda' \rightarrow \Lambda \rightarrow \Lambda'' \rightarrow 0$$

1) If two are in $\mathcal{C}$, so is the third. Use homology exact sequence $\varinjlim$ preserves exactness and "5" Lemma.

2) $\mathcal{C}$ is closed under filtered inductive limits.

Lemma: Any $\mathcal{C}$ with properties (1) and (2) containing $\mathbb{F}_p, \mathbb{Q}$ must be all abelian groups.

Take any abelian group Λ : $0 \rightarrow t\Lambda \rightarrow \Lambda \rightarrow \Lambda/t\Lambda \rightarrow 0$

$$0 \rightarrow \Lambda/t\Lambda \rightarrow \Lambda \otimes_{\mathbb{Z}} \mathbb{Q} \rightarrow \text{torsion group} \rightarrow 0$$

From now on $H_* = H_*(-, \Lambda)$; Λ field. Künneth formula:

$$H_*(X) \otimes H_*(Y) \xrightarrow{\sim} H_*(X \times Y)$$

Consequently for any space X ; $H_*(X)$ is a coalgebra coproduct induced by diagonal $\Delta: X \rightarrow X \times X$.

$$H_*(X) \xrightarrow{\Delta_*} H_*(X \times X) \xrightarrow{\sim} H_*(X) \otimes H_*(X)$$

If X is connected; $H_0(X) = \Lambda$, so $\exists$ distinguished generator of $H_0(X)$ denoted 1.

Have algebra structures on $H_*(GL) = \varinjlim H_*(GL_n)$ induced

Easily seen that $\Delta: GL_p \rightarrow GL_p \times GL_p$ are compatible with $\oplus: GL_p \times GL_q \rightarrow GL_{p+q}$, i.e.

$$\begin{array}{ccc}
 GL_p \times GL_q & \xrightarrow{\oplus} & GL_{p+q} \\
 (\Delta, \Delta) \downarrow & & \downarrow \Delta \\
 GL_p \times GL_p \times GL_q \times GL_q & \xrightarrow{\oplus} & GL_{p+q} \times GL_{p+q} \\
 \alpha \quad \beta \quad \gamma \quad \delta & & (\alpha+\gamma, \beta+\delta)
 \end{array}$$

Commutates

Hence: $\Delta: H_*(GL) \rightarrow H_*(GL \times GL) = H_*(GL) \otimes H_*(GL)$ is an algebra homomorphism $\therefore H_*(GL)$ is a Hopf algebra.

Define $\perp: G_p \times G_q \rightarrow G_{p+q} \quad \begin{pmatrix} 1 & * \\ 0 & \alpha \end{pmatrix} \perp \begin{pmatrix} 1 & *' \\ 0 & \beta \end{pmatrix} = \begin{pmatrix} 1 & * & x' \\ & \alpha & \beta \end{pmatrix}$

Check that this operation induces on these G -matrices induces an algebra structure on $H_*(G_\infty) = \varinjlim H_*(G_n)$ $G_\infty = \bigcup G_n$

$p: G_n \rightarrow GL_n$ is compatible with $\perp$.

$\therefore p_*: H_*(G_\infty) \rightarrow H_*(GL_n)$ is algebra homomorphism

$\therefore s_*: H_*(GL_n) \rightarrow H_*(G_\infty)$ " " " "

Hence $p_* \cdot s_* = 1$ want $s_* \cdot p_* = 1_{H_*(G_n)}$

Lemma: $\begin{pmatrix} 1 & u \\ 0 & \alpha \end{pmatrix} \perp \begin{pmatrix} 1 & u \\ 0 & \alpha \end{pmatrix}$ is conjugate to $\begin{pmatrix} 1 & u \\ 0 & \alpha \end{pmatrix} \perp \begin{pmatrix} 1 & 0 \\ 0 & \alpha \end{pmatrix}$.

Proof:

$$\begin{pmatrix} 1 & u & u \\ 0 & \alpha & 0 \\ 0 & 0 & \alpha \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}}_P \begin{pmatrix} 1 & u & 0 \\ 0 & \alpha & 0 \\ 0 & 0 & \alpha \end{pmatrix} \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}}_{P^{-1}}$$

$$G_p \xrightarrow{\Delta} G_p \times G_p \xrightarrow[1 \times s_p]{1 \times 1} G_p \times G_p \xrightarrow{\oplus} G_{2p}$$

Lemma shows these two homomorphisms are conjugate, so:

$$H_*(G_p) \xrightarrow{\Delta_*} H_*(G_p \times G_p) \approx H_*(G_p) \otimes H_*(G_p) \xrightarrow[\text{id} \otimes s_p]{\text{id} \otimes 1} H_*(G_p) \otimes H_*(G_p) \xrightarrow{\mu} H_*(G_{2p})$$

product

are the same maps. Take limit, replace p by ∞ .

Now we can prove $s_* \cdot p_* = 1$ on $H_n(G_\infty)$ by induction on n .
Assume true for degrees $< n$; let $x \in H_n(G_\infty)$

$$\Delta_* x = 1 \otimes x + \sum_{\deg(x_i') < n} x_i' \otimes x_i''$$

$$\mu(\text{id} \otimes \text{id})(\Delta_* x) = x + \sum_{\deg(x_i'') < n} x_i' \cdot x_i'' \in H_n(G_\infty).$$

$$\mu(\text{id} \otimes s_* \cdot p_*)(\Delta_* x) = s_* p_* x + \sum_{\deg(x_i'') < n} x_i' \cdot s_* p_* x_i'' \quad \text{by above}$$

$$\Sigma = \Sigma, \text{ so } x = s_* p_* x, \text{ completes induction. } \quad \square$$

General fact about Hopf algebras: If C is a Hopf algebra and A is an algebra, one can make $\text{Hom}(C, A)$ into a monoid.

Given $u, v: C \Rightarrow A$ define convolution:

$$u * v: C \xrightarrow{\Delta} C \otimes C \xrightarrow{u \otimes v} A \otimes A \xrightarrow{\mu} A$$

If C is connected, then the algebra C has an inversion, so this monoid is a group.

$$\text{id}^* \cdot \text{id} = \text{id}^* \cdot s_* p_* \quad \text{if group can cancel}$$

Let $B = \left\{ \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} \in M_2 A \right\} = \begin{pmatrix} A & A \\ 0 & A \end{pmatrix}$.

$$GL_{\frac{1}{2}} B = \left(\begin{array}{c|c} \vdots & \vdots \\ \hline \vdots & \vdots \end{array} \right) \quad \text{conjugate get} \quad \left(\begin{array}{c|c} \vdots & \vdots \\ \hline \vdots & \vdots \end{array} \right)$$

$$GL_n B \approx \left(\begin{array}{c|c} GL_n A & A_n A \\ \hline & GL_n A \end{array} \right)$$

Assertion: $H_n B \approx H_n A \oplus H_n A$.

Corollary: $H_* \left(\begin{pmatrix} GL_r M_n \\ 0 & GL_n \end{pmatrix} \right) \leftarrow H_* \left(\begin{pmatrix} GL_r & 0 \\ 0 & GL_n \end{pmatrix} \right)$

induced by inclusion is isomorphism, in the limit as $n \rightarrow \infty$.

Proof:
$$\begin{array}{ccccccc} 1 & \rightarrow & G_n & \rightarrow & \begin{pmatrix} GL_r & M_n \\ & GL_n \end{pmatrix} & \rightarrow & GL_r \rightarrow 1 \\ & & \cup & & \cup & & \parallel \\ 1 & \rightarrow & GL_n & \rightarrow & \begin{pmatrix} GL_r & 0 \\ 0 & GL_n \end{pmatrix} & \rightarrow & GL_r \rightarrow 1 \end{array}$$

Write down group cohomology spectral sequence:

$$E_{pq}^2 = H_p(GL_r, H_q^{\cup}(G_n)) \Rightarrow H_* \left(\begin{pmatrix} GL_r M_n \\ 0 & GL_n \end{pmatrix} \right)$$

$$E_{pq}^2 = H_p(GL_r, H_q^{\cup}(GL_n)) \Rightarrow H_* \left(\begin{pmatrix} GL_r & 0 \\ 0 & GL_n \end{pmatrix} \right)$$

limits preserve exactness.

$\Rightarrow$ isomorphism desired, by Comparison Thm.

Corollary: Now let $r \rightarrow \infty$. $H_* \left(\bigcup_n \begin{pmatrix} GL_r & M_n \\ & GL_n \end{pmatrix} \right) \approx H_* \left(\begin{pmatrix} GL & 0 \\ 0 & GL \end{pmatrix} \right)$.

Thus we obtain embedding: $\begin{pmatrix} A & 0 \\ 0 & A \end{pmatrix} \subset \begin{pmatrix} A & A \\ 0 & A \end{pmatrix} = B$ induces iso

of $H_*(GL(A \times A))$ with $H_*(GL(B))$.

$BGL(A \times A)^+ \rightarrow BGL(B)^+$ is a map of H-spaces
($\Rightarrow$ simple) which is a homology isomorphism, hence a homotopy
equivalence. QED. for assertion.

10/6/74

Thm: A, r fixed

$$H_* \begin{pmatrix} 1 & 0 \\ 0 & GL_r A \end{pmatrix} \hookrightarrow H_* \begin{pmatrix} 1 & M_n A \\ 0 & GL_r A \end{pmatrix} \quad \text{iso as } n \rightarrow \infty.$$

Corollary:

$$H_* \begin{pmatrix} GL_r A & 0 \\ 0 & GL_r A \end{pmatrix} \rightarrow H_* \begin{pmatrix} GL_r A & M_n A \\ 0 & GL_r A \end{pmatrix} \quad \text{as } n \rightarrow \infty.$$
$$H_* \begin{pmatrix} GL_r A & 0 \\ 0 & GL(A) \end{pmatrix} \parallel H_* \begin{pmatrix} GL_r A & GL_{r,n} A \\ 0 & GL(A) \end{pmatrix}$$

Corollary

$$K_* \begin{pmatrix} A & 0 \\ 0 & A \end{pmatrix} \rightarrow K_* \begin{pmatrix} A & A \\ 0 & A \end{pmatrix}$$

$\downarrow$

$$K_* A \oplus K_*(A) \qquad GL_n \begin{pmatrix} A & A \\ 0 & A \end{pmatrix} \approx \begin{pmatrix} GL_r A & M_n A \\ 0 & GL_r A \end{pmatrix}$$

Swan's counterexample:

Recall Milnor's Theorem: If

$$\begin{array}{ccc} D & \xrightarrow{f'} & C \\ g \downarrow & & \downarrow h \\ A & \xrightarrow{f} & B \end{array} \quad \text{is a cartesian (pullback)}$$

square of rings and if either f or g is epic we have
a Mayer-Vietoris sequence.

$$K_1 D \rightarrow K_1 A \oplus K_1 C \rightarrow K_1 B \xrightarrow{\partial} K_0 D \rightarrow K_0 A \oplus K_0 C \rightarrow K_0$$

If both f, g are epic then: can write in the K_2 terms.

$$K_2 D \rightarrow K_2 A \oplus K_2 C \rightarrow K_2 B \xrightarrow{\partial} K_1 D.$$

Swan's Theorem: 2nd conclusion is false if only one is surjective.
 There is no functor $\overline{K}_2$ s.t. for every cartesian square with g ^(split epic) surjective one has:

$$\overline{K}_2 D \rightarrow \overline{K}_2 A \oplus \overline{K}_2 C \rightarrow \overline{K}_2 B \rightarrow K_1 D \rightarrow K_1 A \oplus K_1 C \rightarrow K_1$$

exact.

Example:

$$\begin{array}{ccc} A[\epsilon] & \xrightarrow{f'} & \begin{pmatrix} 0 & A \\ A & A \end{pmatrix} \\ g' \downarrow & & \downarrow g = \text{projection} \\ A & \xrightarrow[\Delta]{f} & \begin{pmatrix} A & 0 \\ 0 & A \end{pmatrix} \end{array} \quad (K_1 = \text{Baso}' K_1)$$

here $A[\epsilon] = A + A\epsilon$ $\epsilon^2 = 0$ = ring of dual numbers / A .

elts: $a + b\epsilon$

$$\begin{aligned} f'(a + b\epsilon) &= \begin{pmatrix} a & b \\ 0 & a \end{pmatrix} \\ g'(a + b\epsilon) &= a \end{aligned}$$

Because g has a section $\Rightarrow \overline{K}_2 C \rightarrow \overline{K}_2 B$, hence:

$$0 \rightarrow K_1(A[\epsilon]) \rightarrow K_1 A \oplus K_1 \begin{pmatrix} A & A \\ 0 & A \end{pmatrix} \rightarrow K_1 \begin{pmatrix} A & 0 \\ 0 & A \end{pmatrix} \rightarrow 0.$$

is exact:

$$\Rightarrow K_1(A[\epsilon]) \underset{g'_*}{\simeq} K_1 A \quad (+)$$

Recall if A is commutative then $K_1 A \simeq A^* \oplus \ker(\det: A \rightarrow A^*)$
 $\simeq A^* \oplus SK_1(A).$

$$0 \rightarrow SK_1 A \rightarrow A \xrightarrow[\det]{\hookrightarrow} A^* \rightarrow 0$$

$$(+) \Rightarrow \begin{array}{l} A[\epsilon]^* \simeq A^* \\ 1 + \epsilon A \\ \text{are units} \end{array} \quad \text{but this is false} \quad \Rightarrow \times$$

Let $\rho: G \rightarrow GL_n A = \text{Aut}(A^n)$.

$$H_* G \xrightarrow{\quad} H_* GL_n A \xrightarrow{\quad} H_* GL(A).$$

Cor:
$$G = \begin{pmatrix} GL_r & M_{rn} \\ 0 & GL_n \end{pmatrix} \xrightleftharpoons[j]{i} GL_{r+n}$$

$$i: \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} \mapsto \begin{pmatrix} a & b \\ 0 & d \end{pmatrix}$$

$$j: \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} \mapsto \begin{pmatrix} a & b \\ 0 & d \end{pmatrix}$$

Then $i_* = j_*: H_* \begin{pmatrix} GL_r & M_{rn} \\ 0 & GL_n \end{pmatrix} \rightarrow H_*(GL)$.

Proof:
$$\begin{pmatrix} GL_r & M_{rn} \\ 0 & GL_n \end{pmatrix} \xrightleftharpoons[j]{i} GL_{r+n}$$

$$\bigcap \begin{pmatrix} GL_r & M_{rn} \\ 0 & GL \end{pmatrix} \xrightleftharpoons[j]{i} \bigcap GL_\infty$$

enough to show $i_* = j_*$ as maps on bottom of square.
But we know:

$$H_* \begin{pmatrix} GL_r & 0 \\ 0 & GL \end{pmatrix} \xrightarrow{\sim} H_* \begin{pmatrix} GL_r & M_{rn} \\ 0 & GL \end{pmatrix}$$

and $i = j$ on $\begin{pmatrix} GL_r & 0 \\ 0 & GL \end{pmatrix}$. QED.

Theorem: F_q = finite field with $q = p^d$ elements, p a prime.

Then:

$$H_i(GL(F_q), \mathbb{Z}/p\mathbb{Z}) = 0 \quad i \geq 1.$$

Proof: Enough to show: $GL_n(\mathbb{F}_p) \hookrightarrow GL(\mathbb{F}_p)$ induces zero map on $\mathbb{Z}_p$ -homology. $\forall n$. Recall that $H_*(P) \rightarrow H_*(G)$ if $[G:P] \not\equiv 0 \pmod{p}$, (transfer argument), and Sylow p -subgp of $GL_n(\mathbb{F}_p)$ is $\begin{pmatrix} 1 & & * \\ & \ddots & \\ 0 & & 1 \end{pmatrix}$.

Enough to show $H_*\left(\begin{pmatrix} 1 & & * \\ & \ddots & \\ 0 & & 1 \end{pmatrix}\right) \rightarrow H_*(GL(\mathbb{F}_p))$ is zero map. $x > 0$

$$\bigcup_{n=1}^T \begin{pmatrix} 1 & * & | \\ & \ddots & | \\ & & 1 & * \\ \hline & & & 1 \end{pmatrix} \subseteq \begin{pmatrix} GL_{n-1} & | & * \\ \hline & & GL_1 \end{pmatrix}$$

$$\begin{matrix} i \\ \downarrow \\ j \end{matrix}$$

$\begin{matrix} i \\ \downarrow \\ j \end{matrix} \leftarrow$ these induce same map on H_* , by corollary.

$\begin{pmatrix} 1 & * & * \\ & \ddots & * \\ & & 1 \end{pmatrix} \subseteq GL$ has same effect as killing last column. Iterate this, use above facts.

Let G be a group. Recall $\mathcal{P}_A$.

By a representation of G over A we mean a pair. (P, ρ) $P \in \mathcal{P}_A$; $G \rightarrow \text{Aut}(P)$ is a homomorphism

Let $\text{Rep}(G, A)$ denote the iso classes of representations of G over A , under $+$ abelian monoid.

Let $S = \text{Iso}(\mathcal{P}_A)$, P_s representative of $s \in S$.

$\text{Hom}(G, \text{Aut}(P))_{\text{Aut}(P)} =$ iso classes of representations of G on projectives $P \approx P_s$. ($P \in \mathcal{P}_A$)

Two representations ρ, ρ' are equivalent if $\theta \rho(g) \theta^{-1} = \rho'(g)$.

$$\text{Rep}(G, A) = \bigsqcup_{s \in S} \text{Hom}(G, \text{Aut}(P_s))_{\text{Aut}(P_s)}$$

Goal: To any representation $E = (P, \rho)$ of G I want a map (canonical).

$$[E]: BG \rightarrow BGLA^+.$$

(as based homotopy class of map).

$$\rho: G \rightarrow \text{Aut}(P) \quad P \oplus Q \cong A^n$$

$$g \mapsto \rho(g) \oplus 1_Q \xrightarrow{\theta} \text{Aut}(A^n) = GL_n(A)$$

Choosing Q, θ we get a homomorphism:

$$BG \rightarrow BGL_n A \rightarrow BGLA \rightarrow BGLA^+.$$

Exercise: Verify independent of choices, well-defined.

Point: $\pi_1(BGL(A)^+) = \pi_1 A$ acts trivially on $BGL(A)^+$, because $BGLA^+$ is an H-space, hence simple.

Properties of: $\text{Rep}(G; A) \rightarrow [BG, BGLA^+]$
 $E \mapsto [E].$

① If T is a trivial representation, $\rho(g) = \text{id}_P \forall g \in G$.
 then $T = (P, \rho)$

② $[E] \oplus [E'] = [E \oplus E']$ $\quad \begin{matrix} [(P, \rho)] = 0\text{-map} \\ \oplus = \text{operation on } [BG, BGLA^+] \\ \text{deduced from H-space structure} \\ \text{of } BGLA^+. \end{matrix}$

③ If $0 \rightarrow E' \rightarrow E \rightarrow E'' \rightarrow 0$ is a s.e.s. of representations.
 then:

$$[E] = [E'] \oplus [E''].$$

Proof of ②:

By adding trivial representations to E, E', E'' we can suppose underlying A -module of E, E', E'' are $\approx A^p, A^q, A^{p+q}$.

$$0 \rightarrow A^p \rightarrow A^{p+q} \rightarrow A^q \rightarrow 0.$$

Can replace G by auto group of this exact sequence:

i.e. $G = \begin{pmatrix} GL_p & M_{q,p} \\ 0 & GL_q \end{pmatrix}$ Have to show:

Enough to show $[E] = [E' \oplus E'']$ assuming ③.

So we want: $G = \begin{pmatrix} GL_p & M_{q,p} \\ 0 & GL_q \end{pmatrix} \xrightarrow[\substack{i \hookrightarrow E \\ j \hookrightarrow E' \oplus E''}]{\quad} GL.$

that $i_* = j_* : BG \rightarrow BGLA^+$.

Enough to show: $p \mapsto \begin{pmatrix} GL_p & M_{q,p} \\ 0 & GL_q \end{pmatrix} \xrightarrow[\substack{i \\ j}]{\quad} GL$ induce same map to $BGLA^+$.

But i, j coincide on subgp: $\begin{pmatrix} GL_p & 0 \\ 0 & GL_q \end{pmatrix}$

and we know that $B \begin{pmatrix} GL_p & \\ & GL_q \end{pmatrix} \xrightarrow{\downarrow} B \begin{pmatrix} GL_p & M_{q,p} \\ & GL_q \end{pmatrix} \rightrightarrows BGL(A)^+$
is homology iso.

Lemma: $X \xrightarrow{f} Y$ homology iso $\Rightarrow M$ H-space connected.
 $[X, M] \xrightarrow{\sim} [Y, M].$

Proof: Puppe: $[X, M] \leftarrow [Y, M] \leftarrow [C_f, M] \leftarrow [\Sigma X, M] \leftarrow \dots$

Show $[C_f, M] = 0$, but Cf is acyclic, so there are no non-trivial maps, by obstruction theory, to any space having no non-trivial perfect subgroups of π_1 .

$$\pi_1 A = \pi_1 (BGL(A)^+)$$

Problem (vague): Describe elements of $[X, BGL(A)^+]$ as some sort of "geometric structures" over X . (analogous to BPL etc)

$\widetilde{KU}(X) = [X, BU] = \varinjlim [X, BU_n] =$ iso classes of n -dimensional complex vector bundles over X a finite complex.

Attempt:

$$\begin{array}{ccc}
 X & \xrightarrow{u} & BGL(A)^+ \\
 \uparrow \text{acyclic} & & \uparrow \text{acyclic} \\
 X' & \rightarrow & BGL(A)
 \end{array}$$

pullback

u thus induces map $X' \rightarrow X$ acyclic together with an element of $[X', BGL(A)] = \text{Hom}(\pi_1 X', GL_n(A))$

If X is finite complex, then:

$$[X, BGL(A)^+] = \varinjlim [X, BGL(A)^+]$$

Assertion: X finite. Then an element of $[X, BGL(A)^+]$ gives rise to

$$\begin{array}{ccc}
 X' & \rightarrow & X \\
 \pi_1 X' & \rightarrow & GL_n A.
 \end{array}$$

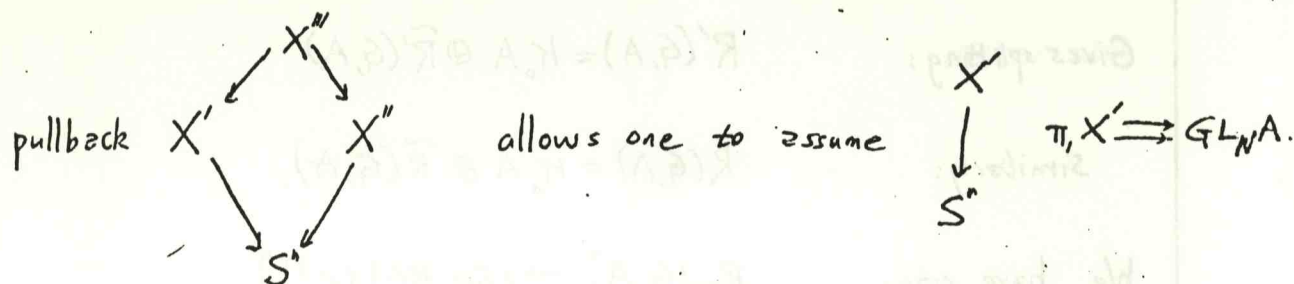
acyclic.

If you have $X' \rightarrow X$ acyclic then: $[X', BGL(A)^+] \cong [X, BGL(A)^+]$

Use the fact that $\pi_1(BGL(A)^+)$ is abelian.
& universal property of acyclic maps. (on killing perfect subgroup)

Conclusion: (For X finite) every pair $(X' \xrightarrow{g} X, \pi_1 X' \rightarrow GL_n A)$ with g acyclic determines an element of $[X, BGL(A)^+]$.
If finite every element of group is obtained this way.

Trouble - I don't know when two pairs give the same element of $[X, BGL(A)^+]$. Case of S^n :



This reduces to the question of when two homomorphisms $\pi, Y \rightarrow GL_N$ determine the same map as $[X', BGL(A)^+]$.

We have seen this is the case if α, β are Jordan-Hölder isomorphic. (This is the theorem: $0 \rightarrow E' \rightarrow E \rightarrow E'' \rightarrow 0$ is s.e.s. of representations of $G \Rightarrow [E] = [E' \oplus E''] \in [BGL(A)^+]$).

Problem: converse.

Recall $Rep(G, A) = \coprod_s Hom(G, Aut(P_s))_{Aut(P_s)}$.

$R(G, A)$ = Grothendieck group of representations with:

$$[E] = [E'] \oplus [E''] \text{ if } 0 \rightarrow E' \rightarrow E \rightarrow E'' \rightarrow 0 \text{ s.e.s.}$$

$R'(G, A)$ = larger Grothendieck with relations:

better notation.

$$R_{\oplus}(G, A).$$

$$[E' \oplus E''] = [E'] + [E''].$$

large use direct sum

= abelian group generated by $Rep(G, A)$.

$$R'(G, A) \twoheadrightarrow R(G, A).$$

$$R'(G, A) \xrightarrow[\text{inclusion}]{\text{forget}} R'(e, A) = \pi_0 A$$

Gives splitting: $R'(G, A) = K_0 A \oplus \tilde{R}'(G, A)$

Similarly: $R(G, A) = K_0 A \oplus \tilde{R}(G, A)$

We have map: $Rep(G, A) \rightarrow [BG, BGL(A)^+]$

$$E \mapsto [E]$$

① trivial repr. go to 0 ② $0 \rightarrow E' \rightarrow E \rightarrow E'' \rightarrow \Rightarrow [E] = [E'] + [E'']$

Lemma: M connected H -space. ($\sim$ CW complex). Then M has a homotopy inverse (so $[X, M]$ is a group.)

$$\begin{array}{ccc} (m_1, m_2) & \xrightarrow{\quad} & (m_1, m_1 m_2) \\ g: M \times M & \xrightarrow[h.equiv]{\quad} & M \times M \\ \downarrow pr_1 & & \downarrow pr_2 \\ M & = & M \end{array}$$

since h.equiv on fibre and base space

g is a map of fibrations, so by long exact homotopy sequence: g is weak homotopy equivalence $\Rightarrow g$ hom. eq.

$$g_*: [X, M] \times [X, M] \rightarrow [X, M] \text{ gives inverse.}$$

By universal property of $R(G, A)$ we get:

$$\tilde{R}(G, A) = R(G, A) / K_0 A \xleftarrow{\quad} R(G, A) \xrightarrow{\quad} [BG, BGL(A)^+]$$

Given any X then we have: $X \rightarrow B\pi_1(X)$

$$\begin{aligned} \text{so you get a canonical map: } \tilde{R}(\pi_1 X, A) &\xrightarrow{\pi^*} [B\pi_1 X, BGL(A)^+] \\ &\rightarrow [X, BGL(A)^+] \end{aligned}$$

Theorem: Let X range over finite complexes. (p -td connected)

Then any natural transformation: $h: \bar{R}(\pi, X, A) \rightarrow [X, Z]$
 with Z having no non-trivial perfect subgrp of π, Z , extends
 uniquely to a nat. trans:

$$\tilde{h}: [X, BGL(A)^+] \rightarrow [X, Z].$$

Example. λ -ring structure on $\bar{R}(\pi, X, A)$ induces operations:

$$\begin{array}{c} \tilde{\lambda}_k \\ [X, BGL(A)^+] \longrightarrow [X, BGL(A)^+]. \end{array}$$

A representation of πX over A may be identified with a
 fibre bundle with fibre ^{class} $\mathbb{P}$ in $\mathbb{P}_A$.

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Theorems as above

Let X range over the category of pointed finite complexes, morphisms
 are homotopy classes of base point preserving maps.

Consider a natural transformation $\gamma: F(\underline{}) \rightarrow G(\underline{})$ F, G map
 to Sets . Say γ has property $(*)$ if:

given $F(X) \xrightarrow{\alpha} [X, Z]$

where Z is a space s.t. π, Z has no nontrivial perfect
 subgroups, then:

$$\begin{array}{ccc} F(X) & \xrightarrow{\alpha} & [X, Z] \\ \gamma(X) \downarrow & \nearrow \exists ! \theta & \\ G(X) & & \end{array}$$

Theorem: The canonical natural transformation

$$[X, BGL(A)] \rightarrow [X, BGL(A)^+]$$

has property (*).

Proof: Fact: If Y is a CW complex and $F_\alpha \subseteq Y$, $\alpha \in J$ is a directed system of finite subcomplexes st. $\bigcup F_\alpha = Y$, then:

$$[X, Y] \xleftarrow{\approx} \varinjlim_J [X, F_\alpha]. \quad X \text{ finite, image caught up in some } F_\alpha$$

Take F_0 to be the 2-skeleton of $B\mathcal{M}_5 \subseteq BGL_5(A)$.

By choosing $B\mathcal{M}_5$ suitably (Milnor model), obtain finite com.

$$\pi_1(F_0) = \pi_1(B\mathcal{M}_5) = \mathcal{M}_5 \text{ perfect group.}$$

By attaching a single 2- & 3-cell to F_0 we obtain $F_0 \hookrightarrow F_0^+$ (go over proof). Form:

$$BGL(A) \cup_{F_0} F_0^+ = BGL(A)^+$$

$$\begin{array}{ccc} F_0 & \xrightarrow{\text{cyclic}} & F_0^+ \\ \downarrow & & \downarrow \\ BGL(A) & \xrightarrow{\text{cyclic}} & BGL(A) \cup_{F_0} F_0^+ \end{array}$$

Van Kampen:

$$\pi_1(BGL(A) \cup_{F_0} F_0^+) = GL(A) / \text{normal subgrp by } \mathcal{M}_5.$$

$\downarrow$
this is $E(A)$

Because it contains $\mathcal{M}_5$ hence gives matrices $\alpha, \beta \in GL(A)$ mod this normal subgrp, α, β commute.

Now with $Y = BGL(A)$, take $F_\alpha =$ all finite subcomplexes of Y containing F_0 .

$$BGL(A) = \bigcup F_\alpha.$$

$$\Rightarrow [X, BGL(A)] = \varinjlim [X, F_\alpha]$$

Hence: $BGL(A)^+ = \bigcup_{F_0} (F_\alpha \cup F_0^+)$

$$\Rightarrow [X, BGL(A)^+] = \varinjlim_{F_0} [X, F_\alpha \cup F_0^+]$$

$$\text{Nat. Trans. } ([X, BGL(A)^+], T(X)) = \varinjlim_{F_0} \text{Nat Transf } ([X, F_\alpha], T)$$

$$= \varinjlim_{F_0} T(F_\alpha) \quad \text{Yoneda's Lemma.}$$

similarly $\text{Nat Trans. } ([X, BGL(A)^+], T(X)) = \varinjlim_{F_0} T(F_\alpha \cup F_0^+)$

Take $T = [, Z]$ so:

$$\text{Nat. Trans. } ([X, BGL(A)^+], [X, Z]) = \varinjlim_{F_0} [F_\alpha \cup F_0^+, Z]$$

$$\downarrow$$

$$\text{Nat. Trans } ([X, BGL(A)], [X, Z]) = \varinjlim_{F_0} [F_\alpha, Z]$$

We know $\pi_1(Z)$ has no nontrivial perfect subgroups. $\Rightarrow$

$$[F_\alpha, Z] \xleftarrow{\alpha} [F_\alpha \cup F_0^+, Z] \quad \text{by acyclicity of map } F_\alpha \rightarrow F_\alpha \cup F_0^+.$$

We assume finite CW complexes since now:

$$X \text{ finite} \Rightarrow \pi_1 X \text{ fin. gen. gp.} \quad [X, BGL(A)] = \text{Hom}(\pi_1 X, GL(A)) = \varinjlim \text{Hom}(\pi_1 X, GL_n A)$$

$$R'(\pi_1 X, A) \twoheadrightarrow R(\pi_1 X, A)$$

$\parallel$
abelian group generated by the monoid of iso classes of rep. of $\pi_1 X$ in $\mathcal{P}(A)$

$$\tilde{R}'(\pi_1 X; A) = \tilde{R}'(X, A) / R'(e, A) = \pi_0 A.$$

Exercise: Show $\tilde{R}'(\pi, X, A) =$ abelian group generated by the mono.
 $\varinjlim \text{Hom}(\pi, X, GL_n A)_{GLA}$
 130 classes of rep. of π, X on A^n .

$$\begin{array}{ccc} \varinjlim \text{Hom}(\pi, X, GL_n A) = [X, BGL(A)] & \longrightarrow & [X, BGL(A)^+] \\ \downarrow & \nearrow h & \nearrow k \\ \oplus \tilde{R}'(\pi, X, A) & & \\ \downarrow & & \\ \tilde{R}''(\pi, X, A) & & \\ \downarrow & & \\ 0 & & \end{array}$$

Theorem: h, k have property (*)

Suffices to show it for h , by diagram chasing.

Fact: Suppose M is an abelian monoid; $\bar{M} =$ abelian group gen.
by M . Grothendieck construction:

$$\begin{array}{ccc} M \times M \times M & \xrightarrow[\cdot]{a, b} & M \times M \xrightarrow{b-a} \bar{M} \\ (m_1, m_2, m_3) & \xrightarrow[\cdot]{a} (m_1 + m_2, m_3) & \text{exact.} \\ & \xrightarrow[\cdot]{b} (m_1, m_2) & \end{array}$$

$$(m'_1, m'_2) \sim (m_1, m_2) \iff \exists t \text{ st. } m'_1 + m_2 + t = m'_2 + m_1 + t.$$

exact means: given $\varphi: M \times M \rightarrow S$ $\varphi a = \varphi b$ then φ factors uniquely thru $\bar{M}$.

$$\begin{array}{ccc} M \times M \times M & \xrightarrow[\cdot]{a, b} & M \times M \xrightarrow{\quad} \bar{M} \\ & \searrow \varphi & \downarrow \varphi \\ & & S \end{array}$$

Proof that $\tilde{h}$ satisfies (*).

Put: $M(X) = \varinjlim \text{Hom}(\pi, X, GL(A))_{GL(A)}$

$$\begin{array}{ccccc}
 M(X) & \xrightarrow[\beta]{\alpha} & M(X)^2 & \rightarrow & \tilde{R}'(X, X; A) \\
 \downarrow u & & \downarrow v & & \downarrow \\
 [X, BGL(A)^+] & \xrightarrow[\beta]{\alpha} & [X, BGL(A)^+]^2 & \rightarrow & [X, BGL(A)^+] \xrightarrow{!} [X, Z] \\
 & & \text{H-space multiplication} & & \text{W.T.}
 \end{array}$$

Both rows exact.

u, v have property (*). (apply previous thm to $A \times A, A \times A \times A$)

Cor from above: $\prod_{i=1}^n [X, BGL(A_i)^+] \rightarrow \prod_{i=1}^n [X, BGL(A_i)^+]$

done now by diagram chases

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$\tilde{K}(X, A) = [X, BGL(A)^+]$ analogous to $\tilde{K}(X) = [X, BU]$

① $A \xrightarrow{u} B$ ring-homomorphism:

$u^*: \mathcal{P}_A \rightarrow \mathcal{P}_B$ $u^*(P) = P \otimes_A B$ also,

$$\begin{array}{ccc}
 \tilde{K}(\pi, X, A) & \rightarrow & \tilde{K}(\pi, X, B) \\
 \downarrow & & \downarrow \\
 \tilde{K}(X; A) & \dashrightarrow & \tilde{K}(X; B)
 \end{array}$$

$A \mapsto \tilde{K}(X, A)$ is covariant functor. $\text{Rings} \rightarrow \mathcal{A}b.$

② transfer: $u: A \rightarrow B$ is a ring homomorphism st. $B \in \mathcal{P}_A$.
Then one has "restriction of scalars".
 $u_*: \mathcal{P}_B \rightarrow \mathcal{P}_A$ $u_*(Q) = Q|_A$.

You get

$$\begin{array}{ccc} \tilde{R}(\pi, X, B) & \longrightarrow & \tilde{R}(\pi, X, A) \\ \downarrow & & \downarrow \\ \tilde{K}(X, B) & \dashrightarrow & \tilde{K}(X, A) \end{array}$$

In particular: $X = S^n$: $u_*: K_n B \rightarrow K_n A$,
called the transfer

③ Products: $K_* A \otimes \tilde{K}(X; A) \rightarrow \tilde{K}(X; A)$, A commutative.

Proposition: (Projection Formula): Suppose $u: A \rightarrow B$ as in (2), $B \in \mathcal{A}$
then:

$$u_* u^*: \tilde{K}(X, A) \rightarrow \tilde{K}(X, A) \quad \text{Forget this.}$$

Remark: If B has a finite projective resolution.

$$0 \rightarrow P_n \rightarrow \dots \rightarrow P_2 \rightarrow P_1 \rightarrow B \rightarrow 0 \quad \text{with } P_i \in \mathcal{A}$$

then one can define a transfer $\tilde{K}(X, B) \rightarrow \tilde{K}_*(X, A)$.
(using other methods).

③ Products again: $\mathcal{P}_A \times \mathcal{P}_B \longrightarrow \mathcal{P}_{A \otimes B}$

$$(P, Q) \longmapsto P \otimes_Z Q.$$

$$\begin{array}{ccc} R(\pi, X, A) \otimes R(\pi, X, B) & \longrightarrow & R(\pi, X, A \otimes B) \\ [E], [F] & \longmapsto & [E \otimes F]. \end{array}$$

Let $\varepsilon: R(\pi, X, A) \rightarrow K_* A$. the map to underlying module
with trivial actions.

$$\begin{array}{c} \downarrow \text{forgets action} \\ R(\pi, X, A) \end{array}$$

Then: $\tilde{R}(\pi, X, A) \otimes \tilde{R}(\pi, X, B) \rightarrow \tilde{R}(\pi, X, A \otimes B)$

$(E - \epsilon E)$ typic element. $(E - \epsilon E) \otimes (F - \epsilon F) = [E \otimes F] - [E] \otimes [F] - [\epsilon F] \otimes E + [\epsilon E] \otimes F$

$$[E], [F] \mapsto [E \otimes F] - [E \otimes \epsilon F] - [\epsilon E \otimes F] + [\epsilon E \otimes \epsilon F] \quad (*)$$

Thus by sending a representation E of π, X over X and a rep. F of π, X over B to above $(*)$ one gets a (representation) natural transformation

$$\begin{array}{ccc} \tilde{R}(\pi, X, A) \otimes \tilde{R}(\pi, X, B) & \xrightarrow{\quad} & \tilde{R}(\pi, X, A \otimes B) \\ \downarrow & & \downarrow \\ \tilde{K}(X, A) \times \tilde{K}(X, B) & \xrightarrow{\quad \mu \quad} & \tilde{K}(X, A \otimes B) \end{array}$$

Properties of μ :

Lemma ① μ is bilinear, associative, commutative.

② Proof: $\mu(\alpha + \beta, \sigma) = \mu(\alpha, \sigma) + \mu(\beta, \sigma)$

$$\tilde{K}(X, A)^2 \times \tilde{K}(X, B) \xrightarrow[\alpha, \beta, \sigma \mapsto \mu(\alpha, \sigma) + \mu(\beta, \sigma)]{\alpha, \beta, \sigma \mapsto \mu(\alpha + \beta, \sigma)} \tilde{K}(X, A \otimes B)$$

$$\tilde{R}(\pi, X, A) \times \tilde{R}(\pi, X, B) \xrightarrow{\quad} \tilde{R}(\pi, X, A \otimes B)$$

Because for $\tilde{R}$ one has $(\alpha + \beta)\sigma = \alpha\sigma + \beta\sigma$. Then same holds for $\tilde{K}$ by uniqueness part of theorem.

associativity: $\tilde{K}(X, A) \times \tilde{K}(X, B) \times \tilde{K}(X, C) \xrightarrow{\quad} \tilde{K}(X, A \otimes B \otimes C)$

$$\tilde{K}(X, B) \times \tilde{K}(X, A) \rightarrow \tilde{K}(X, B \otimes A)$$

$$\parallel$$

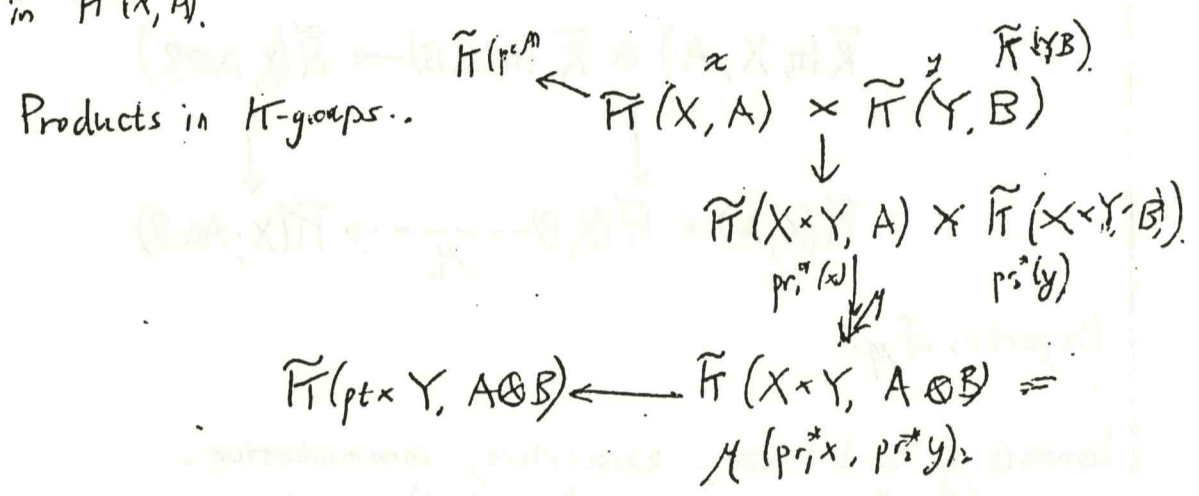
$$\tilde{K}(X, A) \times \tilde{K}(X, B) = \tilde{K}(X, A \otimes B)$$

Can also define products: $\pi_A \otimes \tilde{\pi}(X, B) \rightarrow \tilde{\pi}(X, A \otimes B)$
 $\tilde{\pi}(X, A) \otimes \tilde{\pi}(B) \rightarrow \tilde{\pi}(A \otimes B)$

so that putting $\pi(X, A) = \pi_0 A \times \tilde{\pi}(X, A)$
 one gets products: $\pi(X, A) \times \pi(X, B) \rightarrow \pi(X, A \otimes B)$.

which are associative, commutative, unitary.

Take A to be commutative, so we have $A \otimes A \xrightarrow{\sim} A$, $a \otimes b \mapsto ab$
 Then $\tilde{\pi}(X, A)$ is commutative ring and $\tilde{\pi}(X, A)$ is an ideal in $\tilde{\pi}(X, A)$.



$\mathcal{M}(p_1^*x, p_2^*y)$ dies on $x \times Y$ and $X \times *$ on $X \times Y$, $X \vee Y$

In general.

$$0 \rightarrow [X \wedge Y, Z] \rightarrow [X \times Y, Z] \rightarrow [X \vee Y, Z] \rightarrow 0$$

$\nwarrow$ splitting $\rightarrow [X, Z] \times [Y, Z]$

for H-space Z .

Conclude $\mathcal{M}(p_1^*x, p_2^*y) \in \tilde{\pi}(X \wedge Y, A \otimes B)$
 = subgp of $\tilde{\pi}(X \times Y, A \otimes B)$
 dying on $X \times pt \cup pt \times Y$

So one gets natural transformation:

$$\tilde{\pi}(X, A) \otimes \tilde{\pi}(Y, B) \rightarrow \tilde{\pi}(X \wedge Y, A \otimes B)$$

Take $X = \mathbb{S}^p$, $Y = \mathbb{S}^q$, $X \wedge Y = \mathbb{S}^{p+q}$ and set $\pi(A \otimes B) \rightarrow \pi(A \otimes B)$

Exercise: extend this to p or $q=0$. & check following properties:

- ① associative ② commutative.

$$\begin{array}{ccc} \pi_p A \otimes \pi_q B & \longrightarrow & \pi_{p+q} (B \otimes A) \\ \uparrow \parallel & & \uparrow \parallel \\ \pi_q B \otimes \pi_p A & \longrightarrow & \pi_{p+q} (B \otimes A) \end{array} \quad (-1)^{pq}$$

A commutative $\Rightarrow K_* A$ graded anticommutative ring.

λ -operations and Adams operations in $K_* A$.

Suppose A commutative.

Recall (Atiyah: "K-Theory") :

$$\begin{array}{ccc} K(X) = [X, \mathbb{Z} \times BU] & \begin{array}{c} E \\ \downarrow \\ \Lambda^k E \end{array} & \begin{array}{c} R(A) \\ \downarrow \\ R(A) \end{array} \quad E \mapsto \Lambda^k E. \end{array}$$

one proves this operation induces on bundles operations

$$\lambda^k: K(X) \rightarrow K(X)$$

satisfying: let t be indeterminate $\lambda_t x = 1 + (\lambda^1 x)t + (\lambda^2 x)t^2 + \dots \in K(X)[[t]]$

$$\Rightarrow \textcircled{1} \quad \lambda_t(x+y) = \lambda_t(x) \cdot \lambda_t(y)$$

$$\textcircled{2} \quad \lambda^k(xy) = P_k(\lambda^1 x, \lambda^2 x, \lambda^1 y, \dots, \lambda^k y)$$

$$\textcircled{3} \quad \lambda^k \lambda^l(x) = P_{k,l}(\lambda^1 x, \dots) \quad \text{this is a } \lambda\text{-ring.}$$

Adams Ψ -operations:

$$\lambda_t(L) = 1 + tL$$

L line bundle

$$\text{so } \lambda_t(L) = 1 - tL$$

$$\lambda_t\left(\frac{1}{1-tL}\right)$$

$$= tL + \frac{t^2 L^2}{2} + \frac{t^3 L^3}{3}$$

$$\frac{1}{\lambda_{-t}(L)} = \frac{1}{1-tL}$$

$$\Psi^k L = L^k$$

$$-\ln(\lambda_{-t}(L)) = tL + \frac{t^2 L^2}{2} + \dots$$

$$t \frac{d}{dt} \left(\frac{1}{\lambda_{-t}(L)} \right) = tL + t^2 L^2 + t^3 L^3 + \dots$$

$$\text{Put } \sum_{k \geq 1} t^k \Psi^k x = t \frac{d}{dt} \ln \left(\frac{1}{\lambda_{-t}(x)} \right)$$

$\Psi^0 = \text{id}$ by convention

Identities: (i) $\Psi^k(x+y) = \Psi^k(x) + \Psi^k(y)$

(ii) $\Psi^k(xy) = \Psi^k(x) \cdot \Psi^k(y)$

(iii) $\Psi^k \Psi^l(x) = \Psi^{k+l}(x)$

Theorem: On $\tilde{K}(X, A)$ one has Adams operations, A commutative satisfying (i)-(iii)

$$\tilde{K}(X, A) \otimes \mathbb{Q} = \bigoplus_{n \geq 1} V_n \text{ (eigenspaces)}$$

$$x \in V^n \iff x \in \tilde{K}(X, A) \otimes \mathbb{Q} \quad \text{and} \quad \Psi^k x = \binom{n}{k} x \quad \forall k \geq 1. \quad \text{eigenvalue.}$$

topological analogue: $\tilde{K}(X) \otimes \mathbb{Q} \approx \bigoplus_{n \geq 1} H^{2n}(X, \mathbb{Q})$

? algebraic J-groups,

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$A = \text{commutative ring.}$ $\otimes, \wedge^R$

$R(G, A)$ is a commutative ring with identity with $\lambda^k: R(G, A) \rightarrow R(G, A)$

satisfying:

① $\lambda_t(x+y) = \lambda_t(x) \cdot \lambda_t(y)$
 $\lambda_t(0) = 1$

$$\lambda_t(x) = \sum_{k \geq 0} \lambda^k(x) t^k$$

② $\lambda^k(xy) = P_k(\lambda^1 x, \lambda^k x, \lambda^1 y, \dots, \lambda^k y)$

$$\textcircled{3} \quad \lambda^k(x) = Q_{j,k}(\lambda^1_x, \dots, \lambda^{ik}_x)$$

$$\textcircled{4} \quad L \text{ 1-dimensional } \lambda_t(L) = 1 + t[L]$$

Procedure for calculating P_k $X_1, \dots, X_n, Y_1, \dots, Y_m$.

$$\begin{aligned} \textcircled{2} \quad \lambda^k(xy) &\leftrightarrow k^{\text{th}} \text{ elementary symmetric function of } X_i \times Y_j. \text{ (n^2 vars)} \\ &\quad 1 \leq i \leq n, \quad 1 \leq j \leq m. \\ &= P_k(\text{elem. sym. fns of } X_i, \text{ elem. sym. fns of } Y_j) \\ P_k(\lambda^1_x, \lambda^k_x, \lambda^1_y, \lambda^k_y) &\leftrightarrow \end{aligned}$$

$$\textcircled{2} \quad \lambda^k(x) \leftrightarrow k^{\text{th}} \text{ elementary symmetric function of } X_i \text{'s}$$

$$\sum_{1 \leq i_1 < \dots < i_k \leq n} X_{i_1} \dots X_{i_k}$$

Take j^{th} symmetric function of $X_{i_1} \dots X_{i_k}$ $1 \leq i_1 < \dots < i_k \leq n$ and write it as a polynomial in the elem. sym. functions of X_i .

$$Q_{j,k}(\text{elem. sym. functions of } X_i \text{'s})$$

$$Q_{j,k}(\lambda^1_x, \dots, \lambda^{ik}_x)$$

Adams operations $\Psi^k: R(G, A) \rightarrow R(G, A)$

$$\frac{1}{\lambda_{-t}(x)} = \exp\left(\sum_{m \geq 1} \frac{\Psi^m(x) t^m}{m}\right) \quad \Psi^0 = 1.$$

(looks like Weil zeta function)

1) Ψ is a ring homomorphism

$$\textcircled{3} \quad \Psi^k(\Psi^j x) = \Psi^{kj} x.$$

$$\textcircled{4} \quad \Psi^k \ell = \underbrace{\ell \otimes \dots \otimes \ell}_n \quad \dim \ell = 1.$$

IGA 6: Fact: If A is of characteristic p , then $\exists$ Frobenius endomorphism of A : $Fa = a^p$, which induces map F on $R(G, A)$.

Assertion: $\Psi^p = F$, on $R(G, A)$

γ -operations.

$$x_1^2 + x_2^2 =$$

$$p=2 \quad \Psi^2 x = x^2 - 2\lambda^2 x =$$

$$0 \rightarrow \Lambda^2 P \rightarrow P \otimes P \rightarrow S^2 P \rightarrow 0$$

$$0 \rightarrow K \rightarrow S^2 P \rightarrow \Lambda^2 P \rightarrow 0$$

$$\parallel$$

$$\langle p, p \rangle$$

$$\langle p, p \rangle \rightarrow p \wedge p_2.$$

$$\parallel$$

$$F(P).$$

$$E = L_1 + \dots + L_n. \quad \text{want } \gamma_h(E-n) = k_{\text{elem sym}}^n \text{ for } L_i - 1.$$

$$\text{want } \gamma_t^*(E-n) = \prod (1 + t(L_i - 1)).$$

$$\gamma_t^* = \sum \gamma^h t^h.$$

want γ_t to satisfy:

$$\gamma_t(x+y) = \gamma_t(x) \cdot \gamma_t(y)$$

$$\gamma_t(L-1) = 1 + t(L-1) = (1-t) + tL.$$

$$\parallel$$

$$\gamma_t(L) / \gamma_t(1).$$

$$\gamma_t(L) = (1-t + tL) \gamma_t(1).$$

$$\gamma_t(1) = \frac{1}{1-t}$$

$$= 1 + \left(\frac{t}{1-t}\right)L.$$

$$= \frac{\gamma_t}{(1-t)} L.$$

$$\Rightarrow \boxed{\text{Def: } \gamma_t x = \gamma_{\left(\frac{t}{1-t}\right)}(x)}.$$

 γ -filtration: of $R(G, A)$ is defined by:

$$F_p^\gamma(R(G; A)) = \text{subgp gen. by } \gamma^{i_1}(x_1) \dots \gamma^{i_r}(x_r) \\ i_1 + \dots + i_r \geq p.$$

Fact: On $F_i^\gamma / F_{i+1}^\gamma$ one has $\Psi^k = k^i$.

Fact: In a λ -ring one has:

Ψ^p

Atiyah's book

Theorem: X finite complex. On $\tilde{K}(X, A)$, the γ -filtration is locally nilpotent
i.e. $\forall x \in \tilde{K}(X, A)$, $\exists N_x$ s.t. $\gamma^{i_1}(x) \dots \gamma^{i_r}(x) = 0$ $i_1 + \dots + i_r \geq N_x$.

Corollary: For any $x \in \tilde{K}(X, A)$, $k \geq 1$: $\exists N$ s.t.:

$$(\Psi^k - 1)(\Psi^k - \gamma) \dots (\Psi^k - \gamma^N) x = 0.$$

$\exists$ formula for $(\Psi^k - 1) \dots (\Psi^k - \gamma^N)$ (in terms of γ^i 's of high weight)
in terms of $P(\gamma^1, \dots, \gamma^k)$, involves monomials of weight $\geq N$.

Theorem: A perfect of char. $p \Rightarrow K_0 A$ uniquely p -divisible, i.e.

Pf: Ψ^p is an automorphism on $\tilde{K}(X, A)$ (because $\Psi^p = \text{Frob.}$)

Lemma: If A is an abelian gp with auto. Ψ satisfying:
 $A = \bigcup_n \ker(\Psi - p) \dots (\Psi - p^n)$ then $p: A \xrightarrow{\sim} A$

$$X_n = \ker(\Psi - p) \dots (\Psi - p^n)$$

seen to need:

$$A^{\Psi} \xrightarrow[p]{\sim} A^{\Psi}$$

$$0 \rightarrow X_{n+1} \rightarrow X_n \rightarrow X_n / X_{n-1} \rightarrow 0$$

$$\begin{array}{ccc} \downarrow \Psi & \downarrow \Psi & \downarrow \Psi = p^n \\ 0 \rightarrow X_{n+1} & \rightarrow X_n & \rightarrow X_n / X_{n-1} \rightarrow 0 \end{array}$$

$$0 \rightarrow X_{n+1} \rightarrow X_n \rightarrow X_n / X_{n-1} \rightarrow 0$$

use induction
take union.

$$x \in \tilde{K}(X, A) = \varinjlim [X, BGL_n(A)^+]$$

say $x \in [X, BGL_n(A)^+]$

$$\lambda^k(x+n) = 0 \quad k > n.$$

proof next time.

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A commutative ring.

λ operations on $\tilde{K}(X; A)$.

$$\log\left(\frac{1}{\lambda_{-t}(x)}\right) = \sum_{m \geq 1} \psi_m(x) \frac{t^m}{m}.$$

$$= -\log(\lambda_{-t}(x))$$

Differentiate this:

$$\frac{1}{\lambda_{-t}(x)} (\lambda'_{-t}(x)) = \sum_{m \geq 1} \psi_m(x) t^{m-1}.$$

$$\lambda'_{-t}(x) = \lambda_{-t}(x) \cdot \sum_{m \geq 1} \psi_m(x) t^{m-1}$$

$$\lambda_t(x) = 1 + tx + t^2 \lambda^2 x -$$

$$\lambda'_t(x) = x + 2t \lambda^2 x + \dots \quad ||$$

$$\lambda'_{-t}(x) = x - 2t \lambda^2 x + 3t^2 \lambda^3 x -$$

$$= (1 - tx + t^2 \lambda^2 x - \dots) (\psi'_1 x t^0 + \psi'_2 x t^1 + \dots)$$

$$\psi'_1 x = x$$

$$-2 \lambda^2 x = \psi'_2 x - x \psi'_1 x$$

$$+3 \lambda^3 x = \psi'_3 x - x \psi'_2 x + \lambda^2 x \cdot x.$$

$$-4 \lambda^4 x = \psi'_4 x - x \psi'_3 x + \lambda^3 x \cdot \psi'_2 x - \lambda^3 x \cdot x.$$

Newton $(-1)^{k-1} \lambda^k x = \psi'_k x - x \psi'_{k-1} x + \lambda^2 x \cdot \psi'_{k-2} x - \dots + (-1)^{k-1} \lambda^{k-1} x \cdot x.$

think of $\lambda^k x = \text{det}_{i_1 < \dots < i_k} X_{i_1} \dots X_{i_k}; \quad \psi'_k x = \sum x_i^k.$

Theorem: A perfect of char. $p \Rightarrow K_0 A$ uniquely p -divisible; $p: K_0 A \xrightarrow{\sim} K_0 A$ $n \geq 1$

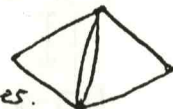
Proof: (1) $\Psi^p = \text{Frobenius}$.

Hence Ψ^1 is isomorphism on $\tilde{K}(X, A)$, for any X . Let $X = S^n = SS^n$

(2) cup-products in $\tilde{K}(SY; A)$ vanish.

general fact about multiplicative cohomology theories.

drag cocycles to diff. sides.



Hence $(-1)^{p-1} p \cdot \lambda^p(x) = \Psi^p(x)$ in $\tilde{K}(SY; A)$. QED.

λ^p is homo now since cup-products vanish.

$$\gamma_t(x) = 1 + t\gamma'_1 x + t^2\gamma''_2 x^2 + \dots = \lambda_{\frac{t}{1-t}}(x).$$

$$\gamma_t(L) = 1 + \left(\frac{t}{1-t}\right)L.$$

$$\gamma_t(L-1) = \frac{1 + \frac{t}{1-t}L}{1 + \frac{t}{1-t}} = 1 + t(L-1).$$

Thm: $\forall x \in \tilde{K}(X, A), \exists n$ s.t. $\gamma^{i_1}(x) \dots \gamma^{i_r}(x) = 0$ $i_1 + \dots + i_r \geq n$.

Cor: $\forall x \in \tilde{K}(X, A), \exists n$ s.t. $\prod_{i=1}^n (\Psi^k - k^i)(x) = 0$.

Because formally one has identities: $\left[\prod_{i=1}^n (\Psi^k - k^i) \right](x) \equiv 0$ mod ideal generated by $\gamma^{i_1}(x) \dots \gamma^{i_r}(x)$, $i_1 + \dots + i_r \geq m$.

Case $k=2$: $\Psi^k L = L^k = (1 + (L-1))^k = 1 + k(L-1) + \binom{k}{2}(L-1)^2 + \dots$

$$\begin{aligned} \Psi^k \left[\sum (L_i - 1) \right] &= \sum \Psi^k (L_i - 1) = \sum (\Psi^k L_i - 1) \\ &= k \left[\sum (L_i - 1) + \binom{k}{2} \sum (L_i - 1)^2 + \dots \right]. \end{aligned}$$

Proof of Theorem: First step is to show $\gamma_t(x)$ is a polynomial in t .

$$\tilde{K}(X, A) = [X, BGL(A)^+].$$

i.e. $\gamma^k(x) = 0$, n large.

$$\text{Hom}(\pi, X, GL_n(A)) = [X, BGL_n(A)] \xrightarrow{\textcircled{1}} [X, BGL_n(A)^+]$$

$$\begin{array}{c} \downarrow \\ [E]^{-n} \\ [A^n] \end{array} \quad \begin{array}{c} \downarrow \\ \tilde{R}(\pi, X; A) \end{array} \xrightarrow{\cong} [X, BGL(A)^+] = \tilde{K}(X, A) \xrightarrow{\gamma^h} [X, BGL(A)^+]$$

$\tilde{R}(\pi, X; A) \xrightarrow{\downarrow \gamma^h} [X, BGL(A)^+]$

Lemma: If $x \in \tilde{K}(X, A)$ comes from $[X, BGL(A)^+]$, then $\gamma^k x = 0$, $k > n$.

Proof: "Universal property" theorem shows the nat. trans. $\gamma^{k \circ \textcircled{2}}$ is determined by $\gamma^h \circ \textcircled{2} \circ \textcircled{1}$.

It suffices to show that: $\gamma^h([E] - n) = 0$ in $\tilde{R}(\pi, X; A)$.

if E is a representation on A^n , for $k > n$.

$$\begin{aligned} \gamma_t(1) &= \frac{1}{1-t}, & \gamma_t([E] - n) &= \gamma_t(E) / \gamma_t(n) = \gamma_t(E) (1-t)^n \\ & & &= \frac{\lambda_t(E)}{1-t} (1-t)^n = \end{aligned}$$

$$= \left(1 + \lambda^1(E) \frac{t}{1-t} + \dots + \lambda^n(E) \left(\frac{t}{1-t} \right)^n \right) (1-t)^n.$$

because $\wedge^k E = 0$, for $k > n$, $\dim E = n$

s. $\gamma_t([E] - n)$ is a polynomial of degree $\leq n$.

$$\gamma_t(x) \cdot \gamma_t(-x) = \gamma_t(0) = 1.$$

Lemma: If $\left(\frac{f, a}{1, 1} \right) = 1$, then coefficients of x_i , $i > 0$, are nilpotent.

(65)

γ -filtration is Locally nilpotent

$\Rightarrow$ (as in Atiyah's book) one gets $\tilde{K}(X, A) \otimes \mathbb{Q} \cong \bigoplus_{i \geq 1} V_i$

where $V_i = \{x \in \tilde{K}(X, A) \otimes \mathbb{Q} : \psi^k x = k^i x\}$.

So for algebraic K-groups: $K_n(A) \otimes \mathbb{Q} \cong \bigoplus V_{(i)_n}$.

$$A^* \xrightarrow{\psi^k} K_1 A \quad \psi^k \alpha = k \alpha, \quad \alpha \in A^* \subseteq K_1 A.$$

$$F \text{ field: } \alpha \in K_2 F \quad \psi^k \alpha = k^2 \alpha.$$

Adams operations in algebraic K-groups are not well-understood.

11.3.74

define suspension of ring A , SA s.t. $K_*(SA) = K_{*+1} A$.

Let $\pi: A \rightarrow B$ ring epimorphism.

Consider triples (E, F, α) $E, F \in \mathcal{P}(A)$ $\alpha: \pi E \xrightarrow{\cong} \pi F$
 $\pi E = E / \ker \pi$.

The set of iso classes of such triples is a monoid: M

$$(E, F, \alpha) \oplus (E', F', \alpha') = (E \oplus E', F \oplus F', \alpha \oplus \alpha').$$

Lemma 1: Any triple (E, F, α) in M , has "inverse" (E', F', α') s.t.

$$\exists: (E, F, \alpha) \oplus (E', F', \alpha') = (A^n, A^n, \text{id}_{A^n}). \quad (A, A, \text{id}_A) \text{ is called basic element of } M$$

Proof: Choose E_1, F_1 s.t. $E \oplus E_1 \approx A^m$ $F \oplus F_1 \approx A^m$.

$$\begin{aligned} \pi E_1 \oplus B^m &= \pi E_1 \oplus (\pi F \oplus \pi F_1) \approx \pi E_1 \oplus \pi E \oplus \pi F_1 \quad (\text{using } \alpha) \\ &\approx B^m \oplus \pi F_1. \end{aligned}$$

(2)

Let V be a vector space over K . Then V is a K -module. $\Rightarrow$ (see the Appendix) we have $\tilde{K}(V) \cong \tilde{K}(V) \oplus \tilde{K}(V) = \tilde{K}(V) \oplus \tilde{K}(V)$

$$\text{where } \tilde{K}(V) = \{ \sum_{i=1}^n \tilde{K}(x_i) \mid \sum_{i=1}^n x_i = 0 \}$$

Let V be a vector space over K . Then $\tilde{K}(V) \cong \tilde{K}(V) \oplus \tilde{K}(V)$

$$V^* \cong K^* \oplus K^* \quad \text{where } V^* = \{ \psi \mid \psi(x) = 0 \}$$

$$\text{Let } \psi \in V^* \text{ then } \psi(x) = 0 \quad \forall x \in V$$

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$$\text{Consider the map } \tilde{K}(V) \rightarrow \tilde{K}(V) \oplus \tilde{K}(V) \text{ defined by } \tilde{K}(x) \mapsto (\tilde{K}(x), \tilde{K}(x))$$

The set of all elements of $\tilde{K}(V)$ is a vector space M

$$(\tilde{K}(x), \tilde{K}(x)) + (\tilde{K}(y), \tilde{K}(y)) = (\tilde{K}(x+y), \tilde{K}(x+y))$$

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